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Mapping forest canopy fuel parameters at European scale using spaceborne LiDAR and satellite data

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ABSTRACT

Spatially explicit data on forest canopy fuel parameters provide critical information for wildfire propagation modelling, emission estimations and risk assessment. LiDAR observations enable accurate retrieval of the vertical structure of vegetation, which makes them an excellent alternative for characterising forest fuel structures. In most cases, fuel parameterisation has been based on Airborne Laser Scanning (ALS) observations, which are costly and best suited for local research. Spaceborne LiDAR acquisitions overcome the limited spatiotemporal coverage of airborne systems, as they can cover much wider geographical areas. However, they do not provide continuous geographical data, requiring spatial interpolation methods to obtain wall-to-wall information. We developed a two-step, easily replicable methodology to estimate forest canopy fuel parameters for the entire European territory, based on data from the Global Ecosystem Dynamics Investigation (GEDI) sensor, onboard the International Space Station (ISS).

First, we simulated GEDI pseudo-waveforms from discrete ALS data about forest plots. We then used metrics derived from the GEDI pseudo-waveforms to estimate mean canopy height (Hm), canopy cover (CC) and canopy base height (CBH), for which we used national forest inventory data as reference. The RH80 metric had the strongest correlation with Hm for all fuel types (r = 0.96–0.97, Bias = −0.16-0.30 m, RMSE = 1.53–2.52 m, rRMSE = 13.23–19.75%). A strong correlation was also observed between ALS-CC and GEDI-CC (r = 0.94, Bias = −0.02, RMSE = 0.09, rRMSE = 16.26%), whereas weaker correlations were obtained for CBH estimations based on forest inventory data (r = 0.46, Bias = 0 m, RMSE = 0.89 m, rRMSE = 39.80%). The second stage was to generate wall-to-wall maps for the continent of Europe of canopy fuel parameters at a resolution of 1 km using a spatial interpolation of GEDI-based estimates for within-fuel polygons covered by GEDI footprints. GEDI observations were not available for some of the polygons (mainly Northern latitudes, above 51.6°N). In these cases, the parameters were estimated using random forest regression models based on multispectral and SAR imagery and biophysical variables. Errors were higher than from direct GEDI retrievals, but still within the range of previous results (r = 0.72–0.82, Bias = −0.18-0.29 m, RMSE = 3.63–4.18 m and rRMSE = 28.43–30.66% for Hm; r = 0.82–0.91, Bias = 0, RMSE = 0.07–0.09 and rRMSE = 10.65–14.42% for CC; r = 0.62–0.75, Bias = 0.01–0.02 m, RMSE = 0.60–0.74 m and rRMSE = 19.16–22.93% for CBH). Uncertainty maps for the estimated parameters were provided at the grid level, for which purpose we considered the propagation of individual errors for each step in the methodology. The final outputs, which are publicly available (https://doi.org/10.21950/KTALA8), provide a wall-to-wall estimation for the continent of Europe of three critical parameters for modelling crown fire propagation potential and demonstrate the capacity of GEDI observations to improve the characterisation of fuel models.

1. Introduction

Global warming and human-induced climate change are leading to increases in temperature, aridity, and drought, which in turn lead to more frequent, more severe wildfires (IPCC, 2022; Jones et al., 2022). They constitute a growing threat to the landscape by causing land use

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change (van Wees et al., 2021) and affecting water, air and soil quality (Rawat et al., 2020; Smith et al., 2011; Voulgarakis and Field, 2015). They also cause biodiversity loss (Kelly et al., 2020), the alteration of carbon and water cycles (Li et al., 2018) and societal impacts (Oliveira et al., 2020; Openko et al., 2023). These negative impacts are mainly the result of alterations in natural fire regimes, caused by socioeconomic and climate changes, and detract from the positive impacts of wildfires in the maintenance of certain communities (Lloret and Zedler, 2009).

The most recent global burnt area products estimate that around 5 Mkm^2 are affected annually by fires (Lizundia-Loiola et al., 2022). In Europe, over 40,000 fires occur annually, which some years affect >800,000 ha (Cardoso Castro Rego et al., 2018; San-Miguel-Ayanz et al., 2023). Even so, the annual European forest burnt area is expected to increase, as climate change increases the frequency of heatwaves and droughts, thereby making extreme fires more likely. These extreme events are characterised by large-front, high-energy releases and affect forest crowns, causing serious impacts on human and ecological values (Duane et al., 2021; Tedim et al., 2018). Although, at present, only about 2% of European fires are classified as extreme fires, they cause most fire-related damage (Duane et al., 2021, Zheng et al., 2023). This is especially true in arid regions, where weather conditions affect both fire ignition and propagation, and indirectly alter the state of the vegetation, reducing its humidity and making it more vulnerable to fire (Cardoso Castro Rego et al., 2018).

Forest fire risk and behaviour are dependent upon weather conditions, topography and fuel properties (Chuvieco et al., 2023; Countryman, 1972). At large spatial extents, fire prevention is dependent on fuel management strategies, involving above all the reduction of fuel loads and opening up gaps in the vegetation in the form of firebreaks. Fuels are characterised by the concepts of fuel types and fuel models (Arroyo et al., 2008). Fuel types refer to homogenous categories of vegetation grouped together on the basis of their fire propagation properties. Fuel models provide quantitative parameters for the fuel types, referring either to surface or canopy fuels. These parameters are input into fire propagation and effect models which estimate fire behaviour (speed, flame length, intensity, etc.), and the impacts of fire. Combining fuel type maps and models (Aragoneses and Chuvieco, 2021) is critical for improving fire monitoring and risk assessment (Aragoneses et al., 2023; Arroyo et al., 2008; Chuvieco et al., 2020; Gale et al., 2021).

Certain canopy parameters are particularly useful for describing forest crown fire behaviour and, as a result, are frequently used in fire management and modelling (Gale et al., 2021). The most common are forest canopy height (Hm, i.e. mean height of trees), canopy cover (CC, i.e. the proportion of total ground area covered by the vertical projection of the tree crowns), canopy base height (CBH, i.e. vertical distance from the ground to the base of the continuous living part of the crown), canopy fuel load (i.e. available canopy fuel normally referring to foliar biomass and branches <5 mm; Keane et al., 2005), and canopy bulk density (CBD, i.e. mass of available canopy fuel per canopy volume unit). Of these five parameters, Hm, CC and CBH provide a good spectrum for modelling fuel properties. Hm affects crown fire spread, biomass consumption and greenhouse gas emissions because it is related with microclimatic conditions and fuel load (Albini, 1976). CC affects potential crown fire propagation because it is related to the horizontal continuity of the fuels. CBH determines the potential transition from surface to crown fires (Ferraz et al., 2009). The wide availability of forest data on Hm and CC provides a unique opportunity to further characterise them at a European level (Ruiz-Benito et al., 2020). For CBH, by contrast, the lack of available data makes this task more difficult.

Traditional methods for estimating forest canopy parameters include field surveys, aerial photogrammetry, and ecological modelling. The first two methods are locally oriented, while the latter method has high degrees of uncertainty, particularly when extended over large regions (Arroyo et al., 2008; Chuvieco et al., 2020; Gale et al., 2021). For the last 20 years, canopy parameters have been estimated using remote sensing data previously calibrated from field data. A wide range of sensors have been used, including passive medium-to-high resolution, multispectral imagery and active sensors such as LiDAR and RADAR, or a combination of sensors (see a recent review in Gale et al. (2021)).

Within LiDAR, Airborne Laser Scanning (ALS) is a technology that has demonstrated a high capacity to estimate forest canopy parameters due to its ability to obtain the vertical structure of forest fuels (Gajardo et al., 2014), with multiple examples for Hm (Riaño et al., 2003), CC (Arumäe and Lang, 2017), CBH (Stefanidou et al., 2020) and CBD (García et al., 2018; Riaño et al., 2004). Most previous studies have used high-density ALS observations (> 4 pulses m^{-2}), while relatively few studies have used ground LiDAR measurements (Alonso-Rego et al., 2021; García et al., 2011).

Despite its potential for estimating fuel parameters at large geographical extents, spaceborne LiDAR has rarely been used to estimate canopy fuel parameters. Examples include the pioneer studies based on the Ice, Cloud, and land Elevation Satellite Geoscience Laser Altimetry System sensor (ICESat/GLAS; García et al., 2012) or the Global Ecosystem Dynamics Investigation sensor (GEDI; Leite et al., 2022). A key limitation of current spaceborne LiDAR sensors is that they do not offer spatially continuous data, which means that the discrete estimations must be spatially extrapolated to generate wall-to-wall products (Liu et al., 2021; Pirotti, 2011). The aforementioned studies extrapolated spaceborne LiDAR observations using machine-learning regression and deep-learning models that combined LiDAR and other satellite observations (García et al., 2017, 2018; Nandy et al., 2021).

Using these approaches, wall-to-wall global Hm was successfully retrieved from GLAS (Lefsky, 2010; Simard et al., 2011) and GEDI sensors (Lang et al., 2022; Potapov et al., 2021). CC was also estimated from GLAS (García et al., 2012) and GEDI data (Rishmawi et al., 2021). However, there are few maps on global forest CC. These include one based on GLAS (Tang et al., 2019), which is not calibrated for Europe, and others based on optical imagery such as the Vegetation Continuous Field (DiMiceli et al., 2015) or another for vegetation over 5 m tall for the year 2000 (Hansen et al., 2013). There are also CC products for Europe, although they are not based on spaceborne LiDAR (European Environment Agency, 2020). ALS-based models have typically been used to estimate CBH (Andersen et al., 2005; Chamberlain et al., 2021; Engelstad et al., 2019; Stefanidou et al., 2020) for which spaceborne LiDAR has yet to be applied. None of these studies covered large extents, focusing instead on local scales or even individual trees (Luo et al., 2018), which means that existing CBH models are not applicable to large areas such as the European continent.

Fuel types were mapped and parameterised for the entire territory of Europe by EFFIS in 2000 (European Forest Fire Information System, 2017) in an approach based on the United States Northern Forest Fire Laboratory (NFFL) standard surface fuel classification system and model (Anderson, 1982). A more recent product at 1-km resolution was developed as part of the European FirEUrisk project (https://fireurisk.eu/, last accessed 2 June 2023). This European fuel map uses a hierarchical, scalable fuel classification system specifically adapted for Europe that covers both surface and canopy fuel categories, and therefore, by extension, both surface and crown fires (Aragoneses et al., 2023). It also includes an initial attempt to assign fuel models to each fuel type, based on the Scott and Burgan categories (Scott and Burgan, 2005). These models only refer to surface fuels, where fires most commonly ignite. However, the increasing importance of extreme fires, which affect forest crowns, means that further efforts must be made to improve the retrieval of forest canopy parameters, so improving in turn the modelling of crown propagation conditions.-

The main objective of our study is to develop methods to estimate and map forest canopy fuel parameters for the whole of Europe in order to improve the modelling of crown fire behaviour in extreme fires. The target resolution is 1 km and the main input is GEDI data, as they include observations for most ecosystems across Europe, except for the boreal regions (> 51.6° N). Three parameters were estimated: forest Hm, CC

and CBH. The estimation was based on models calibrated from GEDI pseudo-waveforms. We aimed to develop methods that could be replicable in other regions covered by GEDI observations, in a bid to complement existing efforts to generate global or continental fire risk assessment systems such as the Global Wildfire Information System (Artés et al., 2019).

To achieve our objectives, we followed five specific steps (Fig. 1). First, we estimated forest Hm, CC and CBH using both discrete ALS and forest inventory data as a reference. Second, we simulated spaceborne GEDI pseudo-waveforms from discrete ALS data. Third, we assessed the relationship between the GEDI pseudo-waveforms and the canopy parameters estimated from ALS and forest inventory data. To this end, we selected the GEDI metric and models that were most closely correlated with the reference data, as the best predictors for Hm and CBH, respectively, and assessed the performance of GEDI-estimated CC. Fourth, European wall-to-wall maps of forest canopy parameters were generated using spatial interpolation of GEDI-based estimations in forest fuel polygons containing GEDI footprints and random forest regression models fed with multispectral, SAR (Synthetic Aperture Radar) and biophysical variables to extrapolate estimations for the polygons for which GEDI observations were not available. Fifth, uncertainty maps for each forest canopy fuel parameter were generated.

2. Material and methods

2.1. Study area

The European Commission has reported that Europe suffers frequent fires, averaging over 40,000 events a year. These result in around 350,000 burnt ha a year, although in some years over 800,000 ha have been burnt (Cardoso Castro Rego et al., 2018; San-Miguel-Ayanz et al., 2023). Depending on the year, between 11 and 44% of the European Natura 2000 Network and other protected areas are affected by wildfires (San-Miguel-Ayanz et al., 2021b, 2022, 2023). Agricultural and tree-dominated vegetation ecosystems are the morst affected ones (San-Miguel-Ayanz et al., 2021b, 2022), as they cover the largest areas of Europe (Aragoneses et al., 2023). Although Southern European countries (Portugal, Spain, Italy, Greece and Southern France) have historically been the most affected by fire (85% of the total European burnt area in 2020), fire activity has recently increased in Central and Northern Europe (San-Miguel-Ayanz et al., 2021a). The fire season (the summer months) is becoming longer, and more extreme fires are taking place at the beginning and end of the season and sometimes even outside it (Cardoso Castro Rego et al., 2018).

The study area comprises the forested area of Europe as described in the 1-km resolution FirEUrisk fuel map. This product was developed on the basis of a novel hierarchical and scalable fuel classification system adapted to the particular fuels and climatic conditions prevailing in European forests, considering both surface and canopy forest fuel types. Fuel types were mapped from a combination of land cover and biogeographic datasets, and bioclimatic modelling (Aragoneses et al., 2023). In this map, the forest covers 1.6 Mkm^2 (32.67% of all of Europe). The most extended forest fuel type is the closed needleleaved evergreen forest, followed by the broadleaved deciduous and closed mixed forests (17.59, 9.23 and 4.75% of the total area of Europe, respectively, Aragoneses et al., 2023).

2.2. Obtaining canopy fuel parameters from ALS and forest inventory data

The GEDI models were calibrated using reference data for the three target forest canopy fuel parameters (Hm, CC and CBH) obtained from ALS observations and forest inventory data. The reference data were obtained from 6587 plots located in forested areas with spatially coincident ALS point clouds across seven sites in three countries, with a broad gradient of European fuel and bioclimatic diversity within the area covered by GEDI (Fig. 2, Table 1).

ALS data for the German, Spanish and Slovenian sites were used as reference data for Hm and CC. Field measurements that were spatially coincident with ALS point clouds were used as reference data for CBH. CBH reference data were not available for the German and Slovenian sites. This parameter was therefore obtained from the four Spanish sites, which covered a wide range of different bioclimatic and biophysical conditions. For the German and Slovenian sites, we generated plots

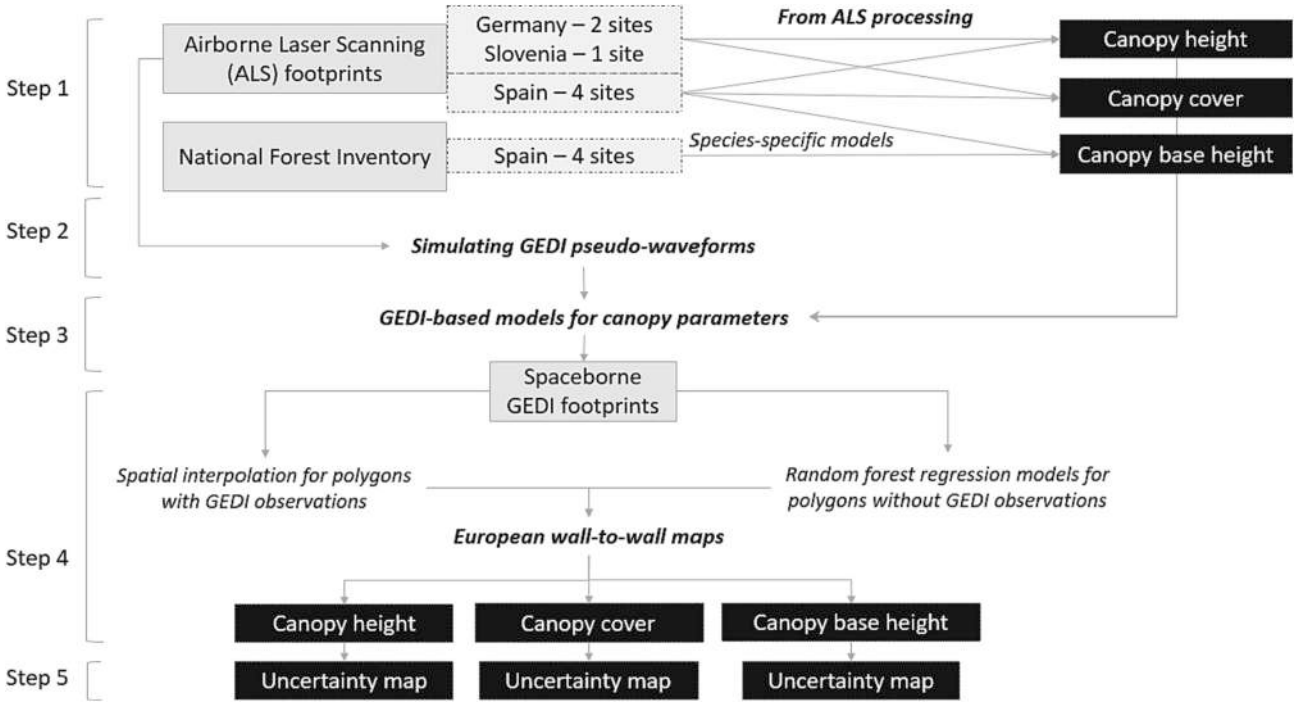


Fig. 1. Flowchart showing the methodology used to generate wall-to-wall maps for forest canopy height, canopy cover and canopy base height at the European level. The databases are shown in grey rectangles, the variables are shown in black rectangles, and the processes are marked in italics.

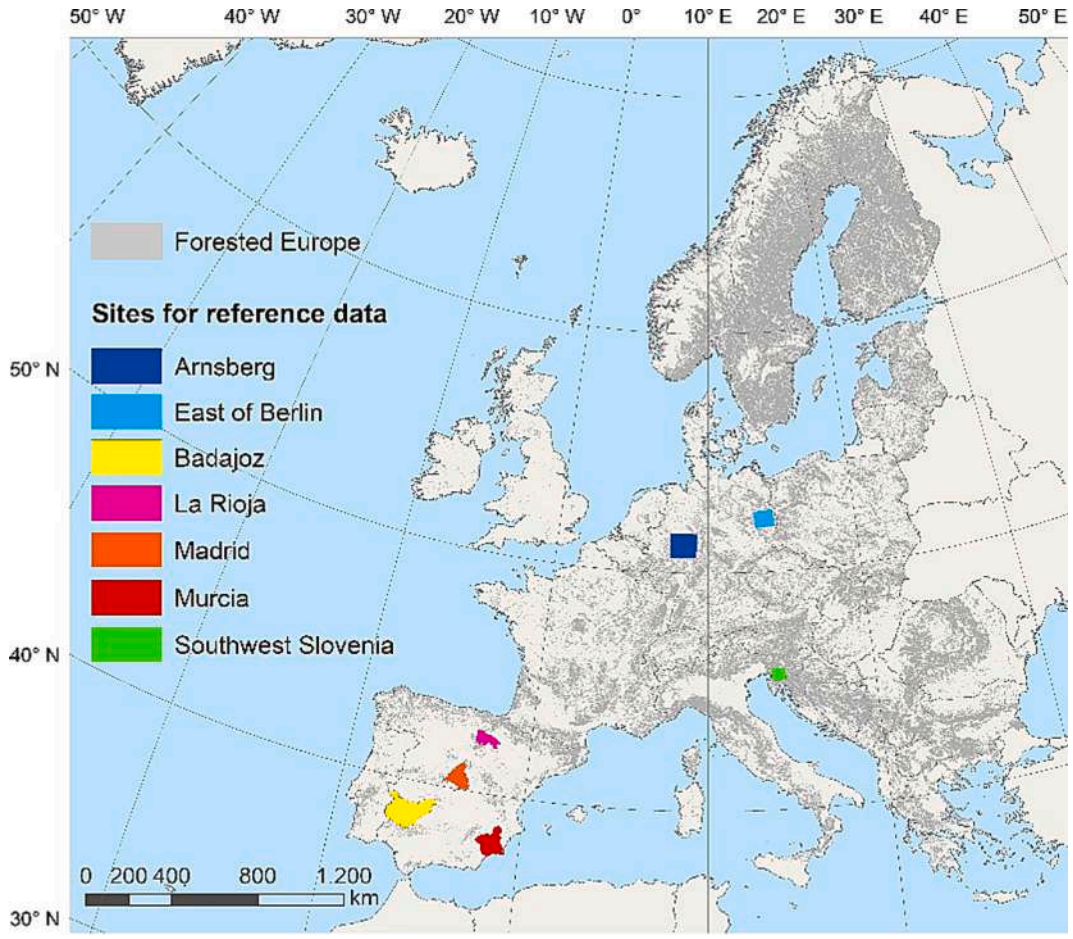


Fig. 2. Reference data used for the calibration of the LiDAR models, overlaid on the forested areas in Europe.

Table 1
Sites used as reference, with climatic data (mean ± standard deviation [monthly min, monthly max]) in the form of annual temperature (T, C°) and annual precipitation (Pp, mm) from 1970 to 2000 WorldClim2 1 km maps (Fick and Hijmans, 2017)), forest fuel types (broadleaved (B), needleleaved (N) and mixed (M)) and ALS information (point density, sampling year and data source).

Site		No. plots	T° (°C)	Pp (mm)	Fuel types	ALS		
						Pts./m^2	Year	Source
Germany	Arnsberg	1339	8.70 ± 1.0 [−4.3, 25.4]	881.38 ± 165.3 [34, 157]	B, N, M	4–10	2016–2021	(GEOportal.NRW, 2023)
	East of Berlin	913	9.04 ± 0.1 [−2.8, 23.7]	546.11 ± 3.1 [30,72]	B, N, M	5	2018–2020	(Laserscandaten, 2023)
Spain	Badajoz	478	15.78 ± 0.6 [0.9, 5.3]	574.64 ± 55.2 [3, 127]	B, N, M	1	2018–2019	(Centro de Descargas del CNIG (IGN), 2023)
	La Rioja	829	10.70 ± 2.1 [6.1, 29.9]	549.55 ± 113.3 [21, 124]	B, N, M	2	2016	
	Madrid	920	12.99 ± 1.9 [−5.1, 34.2]	445.33 ± 136.9 [7, 191]	B, N, M	1	2016	
	Murcia	1032	15.63 ± 1.8 [−4.9, 34.4]	343.90 ± 52.6 [2, 77]	N	0.5	2016–2017	
Slovenia	Southwest Slovenia	1076	9.47 ± 1.8 [−7, 28.3]	1569.24 ± 140.9 [56, 236]	B, N, M	9	2011–2014	(Lidar GIS viewer, 2023)

coincident with the centre of forest pixels in the FirEUrisk European fuel map (Aragoneses et al., 2023). For the Spanish sites, we selected plots from ALS observations that coincided with Spanish National Forest Inventory (NFI) plots (Alberdi et al., 2016) so as to ensure their spatial coincidence with forest cover and their comparability for the calibration of the CBH model.

2.2.1. Forest canopy height and canopy cover from ALS data

We processed the discrete ALS point clouds for the 6587 plots with a diameter of 25 m in order to obtain the forest Hm and CC reference values for each plot. We then created a 2 m Digital Elevation Model (DEM) and a Canopy Height Model (CHM) for each plot after normalizing the height of the returns. To enable comparison with GEDI data, we computed the weighted mean height of the CHM cells emulating the Gaussian distribution of the energy within a GEDI pulse according to the distance to the centre of each footprint. This was done using eq. 1 (Chen, 2010; Zhang et al., 2016).

$$w = e^{\sqrt{\left(\frac{x-x_0}{r}\right)^2+\left(\frac{y-y_0}{r}\right)^2}} \quad (1)$$

where w is the weight of each pixel in the footprint, r is the radius (12.5 m for the circular GEDI footprints of 25 m diameter), x and y are the coordinates of each pixel within the footprint, and x_0 and y_0 are the central coordinates of each footprint.

We also calculated the slope of the footprints from the DEMs using eight neighbours. As regards CC, since forest fuels were defined as trees of ≥2 m in height (Aragoneses et al., 2023), we calculated CC as the number of ALS returns above 2 m divided by all the returns.

2.2.2. Forest canopy base height from national forest inventory data

We estimated CBH for the four Spanish sites with data from the second (1986–1996), third (1997–2007) and fourth (2008–2018) NFI inventories (Alberdi et al., 2016; Villaescusa and Díaz, 1998). The selected sites were located in Badajoz, La Rioja, Madrid and Murcia, covering a gradient of aridity and structural conditions (Tijerín-Triviño et al., 2022). Each NFI plot has a variable radius design, comprised of four concentric circular subplots of 5, 10, 15 and 25 m. In each concentric circle, a selection of adult trees was measured with a diameter at breast height (DBH) of <12.5 cm, < 22.5 cm, < 42.5 cm and ≥ 42.5 cm, respectively. For each sampled tree, we gathered information on the height, DBH and species from the second, third and fourth forest inventory. However, for CBH we had to rely exclusively on the second inventory, as it was the only one that included CBH measurements for a subset of representative species, totalling 25,861 trees (Villaescusa and Díaz, 1998). We used tree height and DBH as predictors for calibrating CBH models per tree species and calculated the evolution of CBH from the year in which the second inventory was carried out to the years for the third and fourth.

From these sampled trees, we selected only those species that had at least 50 individuals with information on CBH. We also selected individuals with a tree height greater than CBH and ≥2 m (i.e. in line with the definition of forest in the FirEUrisk fuel classification system, Aragoneses et al., 2023). We used these data to develop nonlinear mixed-effects models (statistical approaches that combine fixed and random effects) to define the relationship between CBH and other forest parameters for individual trees. We assumed that CBH can be obtained using tree height and DBH using a logistic model (eq. 2) (Fu et al., 2017; Sharma et al., 2017).

$$\mathrm{CBH\ (m)} = \left[\frac{\mathrm{TH}}{1+e^{(b1+b2\ \mathrm{DBH})}}\right] \quad (2)$$

where *TH* is the individual tree height (m), *b1* and *b2* are the parameters to be estimated and *DBH* is diameter at breast height (m).

We fitted mixed-effects models (Zuur et al., 2009) using the nmle package (Pinheiro et al., 2023) in R 4.1.0. (R Core Team, 2021). We considered plot identity as the random effect, assuming that trees within the same plot would be more similar to each other than to trees from the same species in a different plot. This means that the *b1* and *b2* parameters of the equation are different for each plot. We set up control arguments to avoid false convergence due to overparameterisation. We also assessed the goodness of fit through predicted-versus-observed plots for each species, calculating the Pearson linear correlation coefficient (r), the bias (eq. 3), the Root Mean Square Error (RMSE) (eq. 4) and the relative RMSE (rRMSE, eq. 5) (Quinn and Keough, 2002).

$$\mathrm{Bias} = \frac{(\mathrm{Pi}-\mathrm{Oi})}{\mathrm{N}} \quad (3)$$

where P_i is the model-predicted value, O_i is the observed value and N is the number of observations.

$$\mathrm{RMSE} = \sqrt{\frac{\sum(\mathrm{Pi}-\mathrm{Oi})^2}{\mathrm{N}}} \quad (4)$$

where P_i is the model-predicted value, O_i is the observed value and N is the number of observations.

$$\mathrm{rRMSE}\ (\%) = \frac{\mathrm{RMSE}}{\overline{\mathrm{X}}}*100 \quad (5)$$

where *RMSE* is the Root Mean Squared Error (eq. 4) and $\overline{X}$ is the observed mean.

Once we had generated models to estimate CBH for each species in the second NFI, we used tree height and DBH to calculate CBH for all the trees that were alive and growing (i.e. in which CBH was increasing over time) in the second, third and fourth NFI. We calculated the annual increment in CBH by comparing CBH between the third and fourth NFI and added it to the latter to estimate CBH for the ALS year, i.e. the year in which the ALS data for the plot was collected (Table 1). Finally, for each plot we calculated the mean CBH of the trees to obtain the CBH for the plot for its ALS year. To further ensure that CBH was representative of the forested footprints, we only kept plots with at least two trees with estimated CBH, discarding all the others.

2.3. Simulation of GEDI pseudo-waveforms

The GEDI sensor was launched in December 2018 for a 2-year-long mission onboard the International Space Station (ISS). It is the first spaceborne active high-resolution LiDAR mission designed to measure ecosystem structure between 51.6 N and S latitudes, allowing the study of canopy structure, topography, biomass, biodiversity, habitat, carbon and water cycles. The GEDI sensor has three lasers (one coverage laser –"coverage beam"–, and two full-power lasers,–"power beams"–) that obtain full-waveform observations within a 25 m footprint diameter every 60 m along the track and 600 m across the track. It is estimated that it will measure around 4% of the Earth's surface over the predicted course of its mission (Dubayah et al., 2020). GEDI's main products include the geolocated raw waveform (L1B), ground elevation, canopy top height and canopy relative metrics (L2A), canopy cover fraction, profile and leaf area index (L2B), gridded level 2 metrics (L3), footprint level aboveground biomass (L4A) and gridded aboveground biomass density (L4B). GEDI's accuracy when estimating ground elevation and vegetation characteristics has been assessed in various studies (Adam et al., 2020; Dorado-Roda et al., 2021; Liu et al., 2021; Quiros et al., 2021). When compared to other spaceborne LiDAR sensors, such as ICESat/GLAS, GEDI obtained more accurate estimates of Hm (Liu et al., 2021). In our study, ALS and forest inventory data were used as a reference for calibrating models that estimate forest canopy parameters based on the GEDI L2A and L2B products.

The simulation of GEDI-like pseudo-waveforms (Fig. 3) was based on the GEDI simulator (Hancock et al., 2019) and its implementation in the rGEDI package (Silva et al., 2020) in R 4.1.0. (R Core Team, 2021). This approach consists of simulating discrete ALS point clouds to provide similar predicted continuous GEDI-like full waveforms that are expected to have the same properties and errors as GEDI. Although the original GEDI simulator was intended for use in the GEDI mission for calibration and validation purposes, it has been widely used to help estimate forest parameters from ALS data (Dorado-Roda et al., 2021; Nandy et al., 2021). Pseudo-waveforms allow models to be developed without the need for spatially coincident reference data. They also eliminate the effect of geolocation errors, so enabling a better understanding of the relation between GEDI waveforms and the reference data. For all the sites, we used raw discrete ALS point clouds to create GEDI pseudo-waveforms.

We extracted ALS returns for 25 m diameter footprints to match the GEDI footprint size and generate realistic simulations. To this end, a laser footprint width of 5.5 m (1 sigma), which leads to an effective

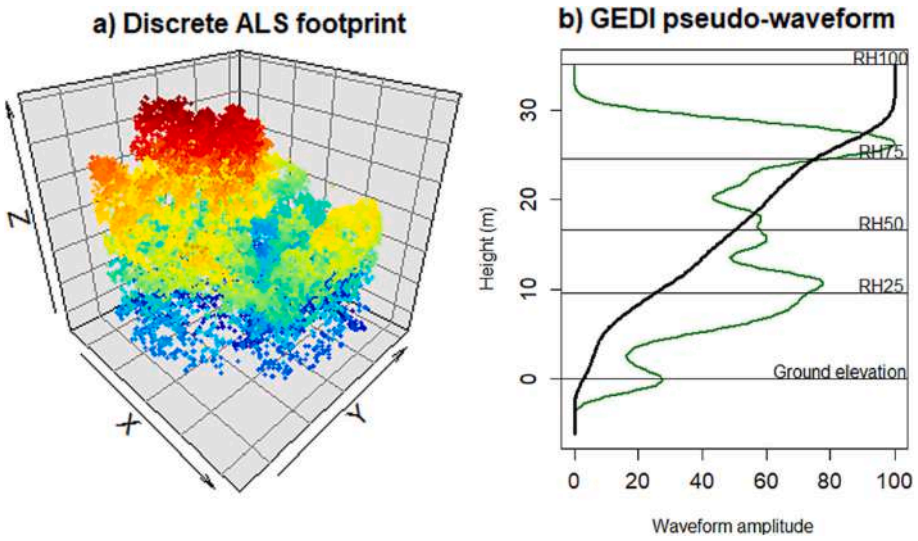


Fig. 3. a) Example of a discrete Airborne Laser Scanning (ALS) point cloud and b) its corresponding GEDI pseudo-waveform. The discrete ALS point cloud includes the 3D point cloud (i.e. longitude, latitude and elevation), whereas the GEDI pseudo-waveform includes the waveform amplitude for each elevation, from which the relative height metrics (RH metrics) can be obtained.

footprint of 22 m, and a Gaussian pulse width as Full Width at Half Maximum of 15 ns were used to match GEDI footprint width and pulse shape. We also derived metrics from the pseudo-waveforms, including Relative Height (RH) metrics, ground elevation, CC and terrain slope (Fig. 3). CC was obtained from the pseudo-waveform following eq. 6.

$$CC = \frac{E_c}{E_c + E_g * \frac{rho_v}{rho_g}} \quad (6)$$

where *CC* is the forest canopy cover obtained from the GEDI pseudo-waveform, *Ec* is the canopy energy, *Eg* is the ground energy, which is calculated as double the energy below ground, and *rho_v* and *rho_g* are correcting factors for vegetation reflectance and ground reflectance, respectively. Although *rho_v* and *rho_g* vary with different types of vegetation and soil, we used generic values of 0.57 and 0.4 respectively, as suggested in rGEDI documentation (Silva et al., 2020).

2.4. Estimation of canopy height, canopy cover and canopy base height from GEDI metrics

We compared the GEDI-estimated RH metrics against the weighted mean of the CHM for each plot to find the RH that was most closely correlated to Hm. The GEDI-estimated CC was compared with the ALS-derived CC to assess the accuracy of GEDI in CC estimation. The GEDI ground elevation, RH metrics, and CC metrics were assessed as candidate predictors for the models estimating CBH.

For forest Hm and CC calibration, we discarded all footprints with a ground location difference of >2 m between the median ALS-derived DEM values and the GEDI pseudo-waveforms, leaving us with just the accurately simulated pseudo-waveforms. For the CC assessment, we applied additional filters to remove the pseudo-waveforms that did not produce coherent CC calculations. These included: 1) footprints whose CC was equal to 0, and 2) footprints whose CC differed by ≥30% for the different methods used to identify the ground elevation of the GEDI pseudo-waveform (i.e. i) CC from the area of the Gaussian-fitted ground; ii) CC from double the energy beneath the Gaussian ground; iii) CC from double the energy beneath the lowest maximum ground; and iv) CC from double the energy beneath the ground at the inflection point). For CBH, we applied some additional filters to help obtain an accurate model for this complex parameter: we only kept footprints where the mean Hm was higher than the reference CBH, and we discarded the plots with a slope of over 20%. We did this to avoid the negative effect of steep slopes when calculating the ground elevation from LiDAR data (Adam et al., 2020), which can strongly influence the estimation of CBH. After applying these filters, the sample size for CBH was reduced from 183 to 103 footprints. In the end, we obtained 5239 plot samples for Hm, 3649 for CC and 103 for CBH.

To retrieve information about the fuel types in each plot, we overlaid the German and Slovenian plots onto the FirEUrisk European fuel map (Aragoneses et al., 2023). For their part, the Spanish plots were superimposed onto the National Forest Map, which had a higher scale of 1:25,000 (Banco de Datos de la Naturaleza, 2022). For Hm, we assessed which GEDI RH metric was most closely correlated with mean canopy height by forest fuel type (i.e. broadleaved, needleleaved and mixed forest). The most closely correlated RH would be used as an estimator of forest Hm. For the CC assessment, we studied the correlation between ALS-derived and GEDI-derived CC for all fuel types.

When it came to selecting the best predictor model for estimating CBH, the sample was too small to allow the generation of different models by fuel types. Instead, we calibrated a linear model using the full sample set, aiming for one parameter for every ten training samples, so as to avoid the risk of overfitting (Harrell et al., 1984). We obtained the best ten independent GEDI predictor metrics (plus the intercept) by exhaustive search. Candidate predictors included the GEDI ground elevation, CC and RH metrics. The best candidate model was compared to other CBH candidate models, discarding one parameter each time according to the Akaike Information Criterion (AIC) (Cavanaugh and Neath, 2019) in order to find the most parsimonious model based on ΔAIC >2 (Burnham and Anderson, 2002). We also calculated the Variance Inflation Factor (VIF) to avoid multicollinearity of the model (James et al., 2013).

The assessment of the best GEDI RH predictor for forest Hm, the correlation of CC and the performance of CBH models were carried out using predicted-versus-observed plots, the Pearson linear correlation coefficient (r), Bias, RMSE, and rRMSE.

2.5. Mapping European wall-to-wall canopy fuel parameters

Once the best GEDI metrics and model had been identified, we then generated continuous maps of forest Hm, CC and CBH for the whole European territory. These maps were primarily based on GEDI footprints. We used GEDI spaceborne L2A and L2B standard products (i.e. footprint-derived metrics and biophysical variables) from Google Earth Engine for spring and summer (May to September) from 2019 to August

2022. We only selected the spring and summer months so as to avoid issues with the footprints of deciduous forests with no leaves in autumn and winter.

We filtered GEDI data according to the accuracy assessment based on the quality, surface and degrade flags (quality flag = 1, surface flag = 1, degrade flag = 0), and discarded outliers in which European forest Hm exceeded 35 m (Adam et al., 2020; Moreno et al., 2017; Pucher et al., 2022). We used both power and coverage beams because coverage beams may only report issues for very tall, dense forests (CC > 95%) (Adam et al., 2020), which are not found in our study area. We then made the selection according to the following criteria: (1) to limit background noise associated with solar radiation, we only selected footprints that had been acquired at night, (2) to ensure good waveform penetration, we used a beam sensitivity threshold of 0.8 (i.e. the CC through which a waveform can penetrate to the ground) as a compromise solution between restricting beam quality and keeping enough footprints for the analysis, (3) to ensure good quality footprints, we discarded those for which GEDI's estimation of ground elevation differed by >2 m from that of the TerraSAR-X DEM (TanDEM-X; German Aerospace Center (DLR), 2018), and (4) to ensure the selection of forest footprints we only kept those that overlapped with both the FirEUrisk forest fuel types (Aragoneses et al., 2023) and the forest cover of the European Space Agency WorldCover Map (Zanaga et al., 2021), which had higher resolution information (10 m) for the whole continent.

To only map canopy fuel parameters for European forest fuel types, we clipped the remaining footprints by forest type using the FirEUrisk European fuel map (Aragoneses et al., 2022). A mosaic of the GEDI-based forest Hm, CC and CBH at 1-km resolution (assessed earlier in section 2.4) was generated as the median for GEDI-derived values for each canopy parameter and reprojected to the ETRS89 Lambert Azimuthal projection. Outliers were removed for forest Hm (2 m < selected RH > 35 m) and CC (CC < 0.15).

We performed a two-step spatial extrapolation to generate the wall-to-wall maps for the Hm, CC and CBH canopy fuel parameters. First, we selected polygons of homogeneous fuel over 1 km^2 within forested cover, as defined by the FirEUrisk fuel type map. These polygons included open and closed broadleaf deciduous forest, open and closed needleleaved evergreen forest, and open and closed mixed forest (Aragoneses et al., 2023). In all the forest fuel polygons containing GEDI footprints, an Inverse Distance Weighted (IDW) interpolation with power = 2.0 was used to estimate fuel parameters for the cells for which no GEDI observations were available. IDW assumes that the influence of the variable decreases in line with the distance to the observations, using a linearly weighted combination of the observed values to determine the values of the unobserved cells (eq. 7). Some of the forest fuel polygons had at least 4 cells with GEDI footprints. 10% of these polygons were set aside so that they could be used later to estimate the uncertainty of the IDW interpolation (see section 2.6).

$$\hat{z}_j = \frac{\sum_i \frac{z_i}{d_{ij}^{\beta}}}{\sum_i \frac{1}{d_{ij}^{\beta}}} \tag{7}$$

where $\hat{z}_j$ is the estimated canopy fuel parameter value at point j and z_i is the fuel parameter value at the ith point. Weights at point i for the calculation of node j are determined by $\frac{1}{d_{ij}^{\beta}}$, where β is the weighting exponent that controls how the weight decreases with distance.

Some fuel polygons did not have GEDI footprints. These were mainly small polygons below 51.6° N and all the polygons above 51.6° N. For these polygons, we developed random forest regression models for each canopy fuel parameter using independent variables as predictors. As independent variables, we used remote sensing images (multispectral and SAR) and biophysical layers that had previously been used in other studies to extrapolate canopy structure parameters, such as Landsat imagery and vegetation indices (Potapov et al., 2021), SAR imagery from ALOS-PALSAR (Advanced Land Observation Satellite-Phased Array type L-band Synthetic Aperture; García et al., 2018; Morin et al., 2019) and climatic and topographic variables (Simard et al., 2011; Zhang et al., 2016; Table 2). We selected atmospherically corrected surface reflectance Landsat-8 images for 2019–2022 to match the year of GEDI footprints, for spring and summer up to August 2022. We masked cloud and shadow pixels and created a Landsat-8 composite based on the medoid, a method for compositing Landsat images resulting in a single observation for every pixel that preserves the relationship between the bands (Flood, 2013). Before extrapolating, we clipped the GEDI-based variables by forest fuel type in the FirEUrisk European fuel map (Aragoneses et al., 2023) and weighted them by the area covered by forest in each 1 km pixel, a value calculated from the FirEUrisk European fuel map (Aragoneses et al., 2023). We reprojected all the canopy fuel parameters to ETRS89 Lambert Azimuthal Equal Area at a resolution of 1 km by averaging the pixel values.

We carried out a stratified random sampling to train the models, using the quantiles of the dependent variables to stratify the samples. Moreover, to ensure that our samples represented sufficiently homogeneous areas, we selected them from the polygons ≥6 km^2 for broadleaved forest, ≥ 4 km^2 for needleleaved forest and ≥ 3 km^2 for mixed forest and discarded the others. The size of the polygons was set so as to ensure that we had at least as many polygons as the number of training samples required for each forest fuel type. Finally, we chose 3500 stratified random samples distributed across the whole of Europe (below 51.6° N to match the GEDI sampling range) representing the full diversity of forest fuels across the continent. 70% of the samples were set aside for training and 30% for validation. The training samples were used to train random forest regression models, an ensemble machine-learning algorithm for estimating continuous variables (Breiman et al., 1984). This algorithm has previously been quite successful at predicting canopy fuel parameters from LiDAR (Healey et al., 2020; Rishmawi et al., 2021). The models used 1000 trees, with five variables per split (number of independent input variables / 3, approximately), one minimum leaf population, a bag fraction of 0.8, no limit to the maximum number of leaf nodes in each tree, and no randomization seed.

The performance of the models was evaluated using the independent validation sample through the Pearson linear correlation coefficient (r), Bias, RMSE, and rRMSE. Random forest also enabled us to determine the importance of each independent variable in the model for each parameter. Finally, we merged interpolated and random forest estimates for all the fuel types (broadleaved, needleleaved and mixed forest) for each of the parameters (forest Hm, CC and CBH).

Once we had generated maps for the canopy fuel parameters by integrating the IDW interpolation and the random forest regression models, we then checked the consistency and coherence of the results. We computed canopy length (CL) as the difference between forest Hm and CBH and used it to check for inconsistencies in the final maps, such as when Hm < CBH. When this happened, we considered the forest Hm values to be more accurate than the CBH values, as forest Hm is directly estimated from LiDAR, whereas CBH is parameterised from plot data at four sites. Therefore, in the pixels where forest Hm was lower than CBH, we recalculated CBH by subtracting the European mean canopy length from the forest Hm by fuel type (broadleaved, needleleaved and mixed forest) for pixels where forest Hm was greater than CBH. After this, if pixels with incoherent CL values (CL < 0 m) continued to appear, they were reclassified to CBH = 0 m.

2.6. Generation of uncertainty maps for the estimation of canopy fuel parameters

Error assessment of the final Hm, CC and CBH maps was performed by generating pixel-level uncertainty maps for each parameter. The errors (RMSE) for each step were assumed to be independent and to propagate to the final estimations as in eq. 8. We considered various different sources of error: the RMSE associated with the calibration of pseudo-waveforms by fuel type, the RMSE associated with spatial

Table 2
Variables used in the regression models to estimate canopy fuel parameters for the fuel polygons for which GEDI observations were unavailable. *: Filtered for 2019–2022.

Dependent variables	Dataset	Target resolution	Reference
Hm (most correlated RH)	From GEDI L2A*	1000 m	Generated in this work
CC	From GEDI L2B*	1000 m	
CBH (best model)	From GEDI L2A*	1000 m	
Independent variables	**Dataset**	**Original resolution**	**Reference**
Spectral data: Blue (482 nm), Green (561 nm), Red (654 nm), NIR (864 nm), SWIR1 (1609 nm), SWIR2 (2201 nm)	Landsat 8 T1_L2*	30 m	(USGS, 2022)
Vegetation spectral indices	Derived from Landsat 8 T1_L2*	30 m	
NDVI (NIR – Red) / (NIR + Red)			(Rouse Jr. et al., 1974)
GNDVI (NIR – Green) / (NIR + Green)			(Gitelson et al., 1996)
EVI 2.5*((NIR – Red) / (NIR + 6*Red – 7.5*Blue +1))			(Huete et al., 1997)
NDMI (NIR – SWIR1) / (NIR + SWIR1)			(Gao, 1996)
NBR (NIR – SWIR2) / (NIR + SWIR2)			(Miller and Thode, 2007)
NBR2 (SWIR1 – SWIR2) / (SWIR1 + SWIR2)			
Median HH and HV polarization backscattering coefficient (gamma nought values)	Global PALSAR-2/ PALSAR yearly mosaic*	25 m	(Shimada et al., 2014)
NDBI: (HH – HV) / (HH + HV)	Derived from Global PALSAR-2/ PALSAR yearly mosaic*	25 m	(Almeida-Filho et al., 2009)
Annual precipitation	WordClim	927.67 m	(Hijmans et al., 2005)
Annual temperature			
Precipitation seasonality			
Temperature seasonality			
Elevation	GTOPO30	927.67 m	(U.S. Geological Survey, 2006)
Slope	Derived from GTOPO30	927.67 m	
Median Digital Surface Model (DSM)	ALOS DSM	30 m	(Takaku et al., 2014)
Median tree cover	MOD44B.006*	250 m	(DiMiceli et al., 2015)
Median net primary production (NPP)	MOD17A3HGF.006*	500 m	(Running and Zhao, 2019)
Top of the canopy height	Canopy height model on Earth based on GEDI RH98	10 m	(Lang et al., 2022)
Biogeographic regions	BiogeoRegions2016	100 m	(European Environment Agency, 2016)

extrapolation (IDW and random forest regression models), and only for CBH, the RMSE associated with the generation of CBH reference data from NFI. For this last source of error, as different errors were obtained for each species, we added the RMSE by fuel type by choosing the broadleaved and needleleaved species with the highest RMSE, and for mixed fuels the highest RMSE for all species. We assumed those RMSE for all broadleaved, needleleaved and mixed pixels, respectively. Finally, the uncertainty at pixel level was computed in a similar way to Silveira et al. (2023): we grouped the predicted parameters into 10 bins according to natural breaks and calculated the RMSE for each bin, taking all the steps into consideration. Next, we fitted linear regression models to predict the RMSE based on the predicted values and the different errors at pixel-level.

$$\mathrm{RMSE}_{\mathrm{final_estimations}} = \sqrt{RMSE_1^2 + \ldots + RMSE_n^2} \quad (8)$$

where *RMSE* refers to the RMSE of each step followed to generate the final maps for Hm, CC and CBH. Depending on the pixel, it can refer to: i) generation of reference data, ii) calibration using the GEDI-like pseudo-waveforms, or iii) spatial extrapolation from IDW or random forest regression models.

3. Results

3.1. Estimation and assessment of canopy fuel parameters

RH80 showed the strongest correlation with ALS-based forest canopy mean height (Hm) for all forest fuel types (Fig. 4; $r = 0.96$–0.97, Bias = −0.16-0.30 m, RMSE = 1.53–2.52 m, rRMSE = 13.23–19.75%), followed by the GEDI RH75 (Figs. A.1–A.3 in Appendix A; r = 0.96, Bias = 1.07–1.20 m, RMSE = 2.47–0.64 m, rRMSE = 14.33–16.66%). RH80 was used to predict Hm for all fuel types at the European level.

The correlation between canopy cover (CC) from the discrete ALS and GEDI pseudo-waveforms showed a strong agreement, with $r = 0.94$, Bias = −0.02, RMSE = 0.09 and rRMSE = 16.26% (Fig. 5). This high level of agreement was obtained after filtering the GEDI pseudo-waveforms (see Fig. A.4 in Appendix A).

A total of nine forest species, including both broadleaved and needleleaved species, had at least 50 trees for which CBH had been measured in plots with at least two trees and the filters in section 2.2. These species were as follows: *Pinus sylvestris, Pinus uncinata, Pinus pinea, Pinus halepensis, Pinus nigra, Pinus pinaster, Quercus pyrenaica, Eucalyptus camaldulensis* and *Fagus sylvatica* (Fig. A.5 in Appendix A). The CBH reference models based on tree height and DBH parameterised with forest inventory data by forest species obtained high accuracy (r = 0.86–0.97, Bias 0.02–0.07 m, RMSE 0.23–0.69 m, rRMSE 13.54–32.65%), with normality of residuals and homogeneity of variance (homoscedasticity).

In the exhaustive search for the best GEDI-based CBH model, the following five metrics were selected: ground elevation (GE), signal start (SE; equivalent to RH0), RH10, RH85 and RH95. However, the final selected model was based on just four metrics, namely GE, SE, RH10 and RH85, as this was the most parsimonious model (i.e. lowest AIC value, eq. 9; Fig. 6; Table A.1 in Appendix A), obtaining $r = 0.46$, Bias = 0 m, RMSE = 0.89 m, and rRMSE = 39.80%. The explanatory variables had VIF values of <3 units (GE = 1.03, SE = 1.78, RH10 = 2.72, and RH85 = 1.88), indicating that there were no multicollinearity problems (James et al., 2013). The final model was checked for linear relationship assumptions and showed normality of residuals and homoscedasticity. Reducing the number of parameters in the model had no effect on its performance (Table A.2 in Appendix A).

The final selected model was eq. 9:

$$\begin{aligned}\mathrm{CBH\ (m)} = & -0.0006515\ \mathrm{GE}^{**} + 0.1958871\ \mathrm{SE}^{*} - 0.2642275\ \mathrm{RH10}^{*} \\ & + 0.1649544\ \mathrm{RH85}^{***} + 2.5386346^{***}\end{aligned} \quad (9)$$

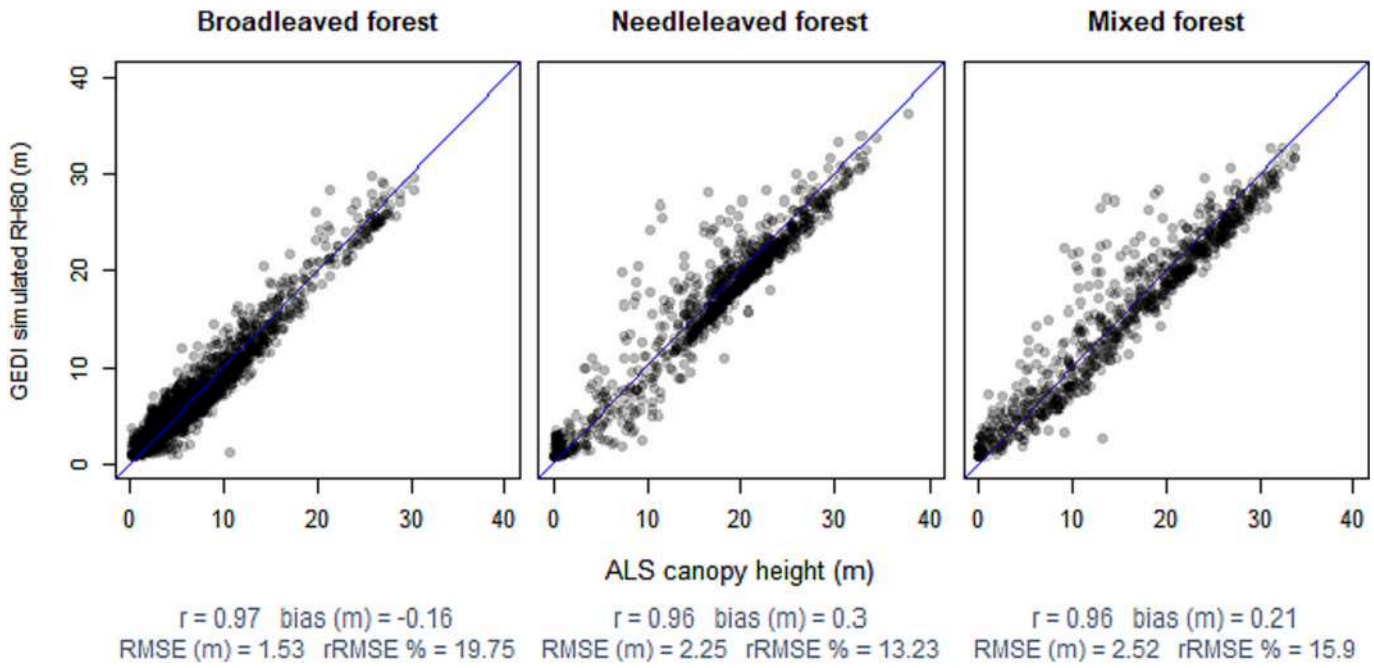


Fig. 4. Simulated GEDI RH80 versus ALS canopy height by fuel type. The blue lines are 1:1. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

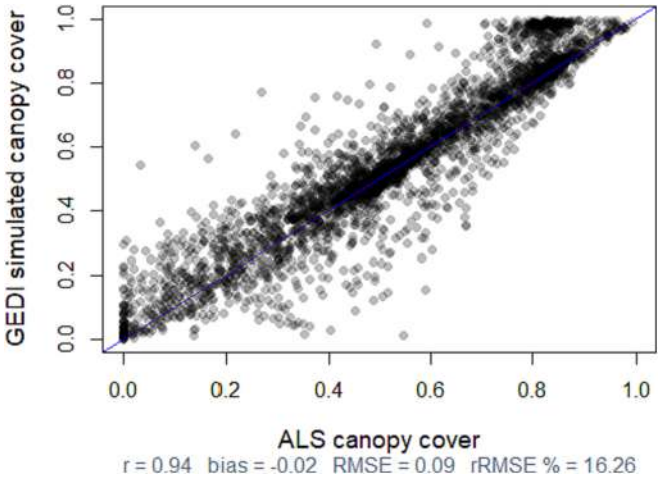


Fig. 5. GEDI-simulated canopy cover versus ALS canopy cover. The blue line is 1:1. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

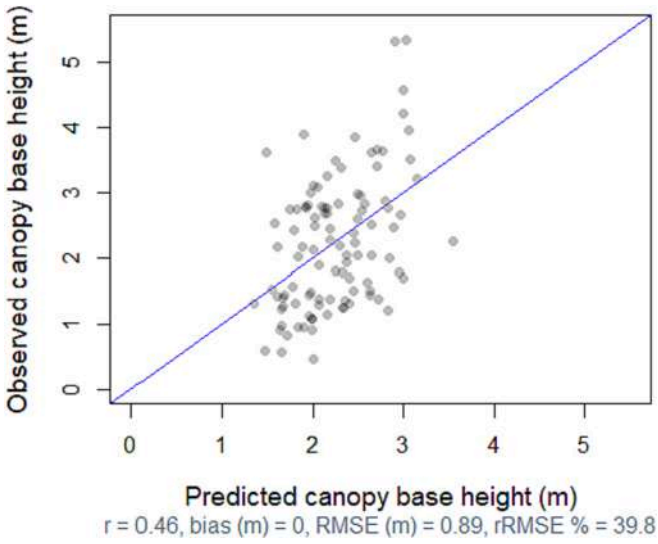


Fig. 6. Predicted versus observed canopy base height values from the Spanish National Forest Inventory estimations and Eq. 9. The blue line is 1:1. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

where *GE* is the ground elevation, *SE* is the minimum elevation (m) the LiDAR signal reaches and *RH* refers to the relative height percentiles at 10 and 85% energy. ***: *p*-value <0.001. **: p-value <0.01. *: p-value <0.05.

3.2. Generation of the European wall-to-wall maps of canopy fuel parameters

We retrieved forest Hm, CC and CBH from the GEDI footprints in the study area using RH80, GEDI CC and the best CBH model, respectively. The forest canopy fuel parameters were interpolated using the IDW method for 47.85% of the European forest, while for the remaining 52.15%, the parameters were estimated on the basis of random forest regression models. Only 8% of the forest located within GEDI coverage was estimated using random forest regression models.

The results of the random forest regression models for Hm by fuel type ranged between r = 0.72–0.82, Bias = −0.18-0.29 m, RMSE = 3.63–4.18 m and rRMSE = 28.43–30.66%. These results were similar for all forest fuel types. Although the model showed high performance, it also displayed a saturation effect above 20 m for all forest types, thus underestimating tall forests. The most important variable to predict Hm for all fuel types was top of the canopy height (i.e. the height of the tallest tree in the pixel), followed by slope and HV polarization from SAR imagery (Fig. 7, A.6.a. in Appendix A).

The random forest regression models for CC also showed good performance by fuel type with r = 0.82–0.91, Bias = 0, RMSE = 0.07–0.09 and rRMSE = 10.65–14.42%. The most important variables for predicting CC were top of the canopy height for broadleaved forests and HV polarization from SAR imagery for needleleaved and mixed forests.

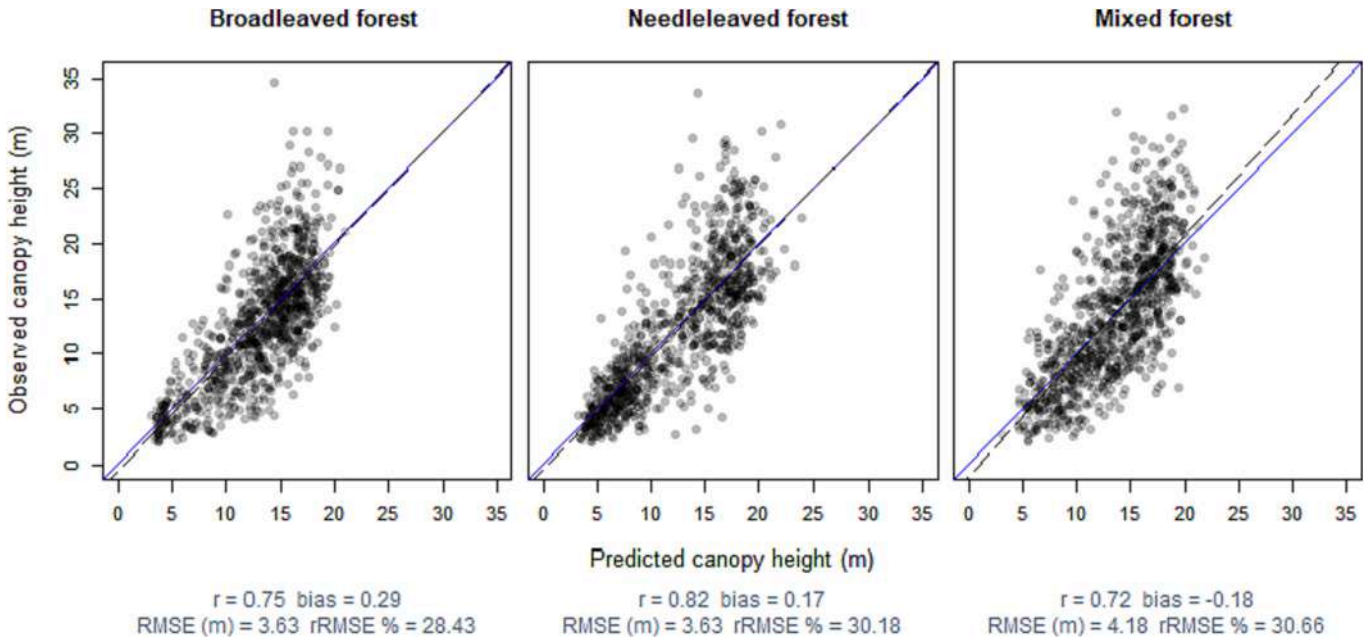


Fig. 7. Predicted versus observed canopy height by fuel type for the random forest regression models. The blue line is 1:1. The dashed line is the best-fit line for our model. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Median tree cover and slope were also important variables for predicting CC for all forest types (Fig. 8, A.6.b. in Appendix A).

As expected, the random forest regression models for CBH did not perform as well as the models for Hm and CC, with $r = 0.62$–0.75, Bias = 0.01–0.02 m, RMSE = 0.60–0.74 m and rRMSE = 19.16–22.93%. The highest RMSE and rRMSE values were for mixed forest. The most important variable for predicting CBH for all fuel types was top of the canopy height (Fig. 9, A.6.c. in Appendix A).

Finally, the combination of GEDI spatially interpolated data and the random forest regression models enabled us to generate the wall-to-wall maps for the distribution of Hm, CC and CBH over European forest fuel types (Fig. 10). Hm was constrained to 2.00–35.00 m with a mean of 9.65 m (standard deviation = 3.37), CC ranged between 15 and 98% with a mean value of 52% (standard deviation = 12%) and CBH varied between 0.00 and 12.54 m with a mean value of 2.90 m (standard deviation = 0.60 m). We found that 0.04% of the estimated CBH values were higher than Hm. In these cases, therefore, the CBH values were corrected by subtracting the mean canopy length of the other 99.96% from their estimated Hm, i.e. 7.27 m for broadleaved forest, 6.48 m for needleleaved forest and 6.94 m for mixed forest. The final 1 km European maps on Hm, CC and CBH are publicly available (Aragoneses et al., 2024) https://doi.org/10.21950/KTALA8.

The uncertainty maps (Fig. 10) show the uncertainty of Hm, CC and CBH estimations as a combination of the error associated with every step involved in the generation of the forest canopy parameters. The highest relative errors tend to appear in mountainous areas and those with

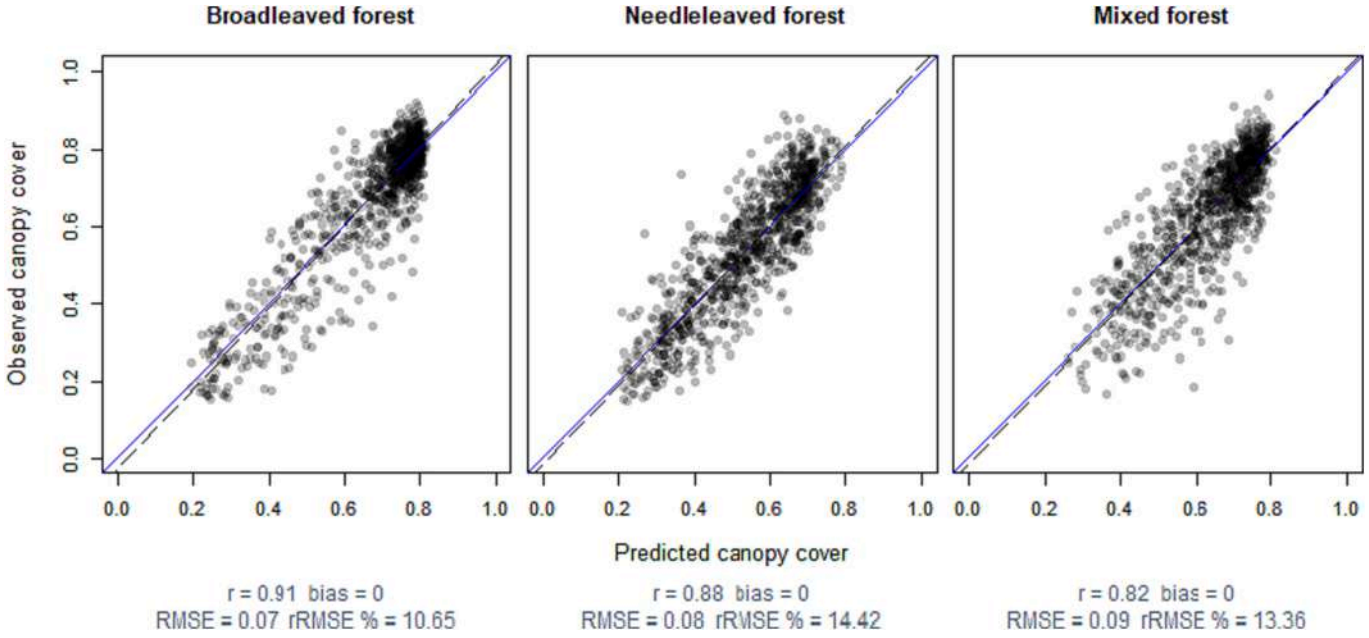


Fig. 8. Predicted versus observed canopy cover by fuel type for the random forest regression models. The blue line is 1:1. The dashed line is the best-fit line for our models. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

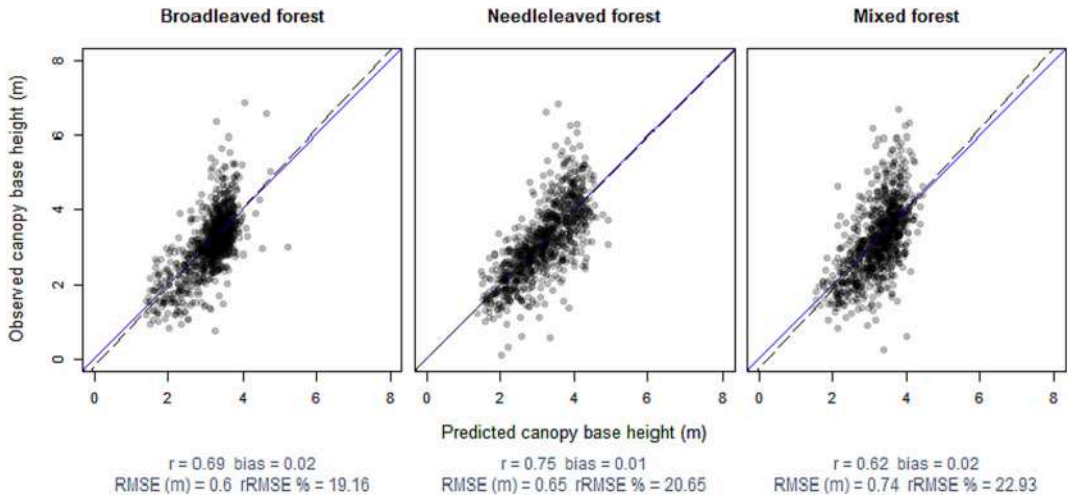


Fig. 9. Predicted versus observed canopy base height by fuel type for the random forest regression models. The blue line is 1:1. The dashed line is the best-fit line for our models. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

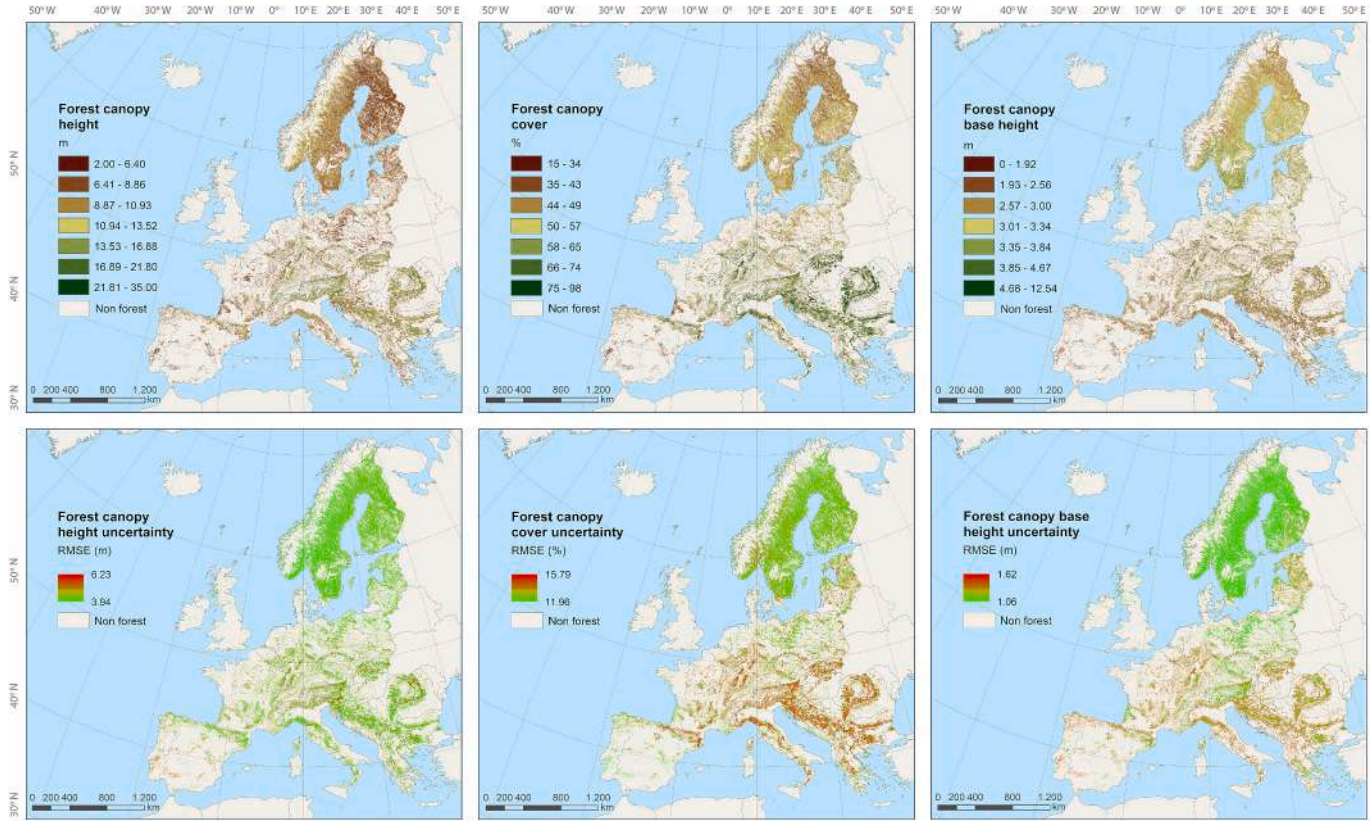


Fig. 10. Maps by Jenks intervals on the distribution of European forest canopy height, canopy cover, and canopy base height, and their pixel-level uncertainty.

mixed fuel type patterns. For Hm, uncertainty ranged between 3.94 and 6.23 m with a mean RMSE of 4.38 m, for CC it ranged between 12 and 16% with a mean RMSE of 14%, and for CBH it ranged between 1.06 and 1.62 m with a mean RMSE of 1.19 m.

4. Discussion

We demonstrated the capacity of GEDI to estimate various forest canopy fuel parameters -forest canopy height (Hm), canopy cover (CC) and canopy base height (CBH)- at a European scale. This analysis first required linking GEDI metrics to forest inventory and ALS reference data by simulating GEDI pseudo-waveforms and then extrapolating this information within the polygons for which GEDI footprints were available. For the polygons that did not have GEDI data, we used random forest regression models based on multispectral and SAR images, and biophysical variables. The final maps combined both sets of data to obtain 1 km resolution wall-to-wall canopy fuel parameters at a European scale, and together with their pixel-level uncertainty maps, they can improve wildfire risk assessment in Europe by enabling better modelling of crown fire risk potential.

The methods proposed in this paper made it possible to estimate canopy fuel parameters at a continental scale, which are critical for fire

behaviour modelling. Our approach involved modelling in two stages, firstly to estimate the reference parameters from GEDI observations and then to spatially extrapolate the resulting estimates to create wall-to-wall products. The compatibility of real and simulated GEDI footprints (Huettermann et al., 2022; Silva et al., 2018) allowed us to assess the capacity of GEDI to estimate canopy fuel parameters at a European level without needing to perform the traditional time- and energy-consuming field campaigns (e.g. Andersen et al., 2005; Erdody and Moskal, 2010; García et al., 2017). Thus, when few independent field datasets for large geographical extents are available, GEDI-simulated pseudo-waveforms can be used to calibrate the models and validate the forest parameters (Huettermann et al., 2022; Silva et al., 2018).

Even though the ALS data we used for simulating the GEDI pseudo-waveforms came from just three countries, we tried to compensate for the limited amount of available data by choosing representative sites covering a wide range of geographical and climate conditions, so as to ensure that we were covering the full diversity of European forest fuel types (Hancock et al., 2019; Huettermann et al., 2022; Marino et al., 2022). In addition, reference data for CBH could also be improved if new systematic CBH measurements spatially coincident with ALS data were available. Future research should focus on increasing the sample size of the ALS and CBH reference data so as to encompass as far as possible the variability of the different European fuel types across a range of climatic, topographic and structural conditions.

GEDI geolocation uncertainty, GEDI $\sim$ N (μ = 0 m, σ = 10 m) (Dubayah et al., 2020) may influence the estimation of fuel parameters, especially at high-resolution scales and in heterogeneous forests (Roy et al., 2021; Wang et al., 2022). However, our parameterisation was not subject to geolocation uncertainty, as the GEDI-simulated pseudo-waveforms were derived from ALS point clouds with lower errors. Moreover, for the spatial extrapolation to Europe of the GEDI spaceborne footprints, the resampling of the GEDI metrics and auxiliary variables to the target spatial resolution of 1 km minimised this source of uncertainty because of the coarser scale used at European level. Moreover, in line with Lang et al. (2022), we filtered out the GEDI spaceborne footprints that did not overlap with forest cover. This reduced uncertainty by removing the footprints that overlapped with other non-forest fuel types.

GEDI RH80 obtained the most accurate results when estimating forest Hm (r = 0.96–0.97, Bias = −0.16-0.30 m, RMSE = 1.53–2.52 m, rRMSE = 13.23–19.75%). Other GEDI-based global-scale research studies have focused on maximum forest height: Potapov et al. (2021) found GEDI RH95 to be the closest metric to the overall maximum height of tree canopies (RH95 r = 0.71); while Lang et al. (2022) and Healey et al. (2020) directly mapped the top of the canopy height for the forest, defined as GEDI RH98. However, none of these studies calibrated forest Hm for Europe, which has differences in forest height and density compared to other forested regions of the world. Our final map cannot strictly be compared to these studies because they used a different definition of forest Hm. We obtained an estimated Hm from 2.00 to 35.00 m with a mean of 9.65 m, which is in line with the mean height of temperate and boreal European forests and somewhat shorter than that of tropical forests (Potapov et al., 2021).

For the forest CC parameter, we obtained a strong correlation between GEDI and ALS CC (r = 0.94, Bias = −0.02, RMSE = 0.09, rRMSE = 16.26%). This was greater than the correlation obtained for global forest CC with other spaceborne LiDAR sensors, for example between GLAS and LVIS (Land, Vegetation and Ice Sensor) (r = 0.27), and GLAS and MODIS (Moderate Resolution Imaging Spectroradiometer) Vegetation Continuous Field (r = 0.29) (Tang et al., 2019). The filtering process we applied seems to have been particularly effective (see Figs. 5 and A.4 in Appendix A), especially for steep slopes, where large differences were found between the different methods used to calculate ground elevation.

One limitation of this research is that for the CBH parameter, only the Spanish sites could be used. This could be improved in the future if new systematic forest inventories include CBH or if specific field campaigns are carried out for this purpose across Europe. However, the Spanish sites represent a wide variety of structural conditions in forest and bioclimatic diversity, which allow the results to be extrapolated to larger areas. We obtained highly accurate models for different species, representative of common European species and fuel types (Köble and Seufert, 2001). The best-performing model for reference forest CBH was built with 4 GEDI metrics (r = 0.46, Bias = 0 m, RMSE = 0.89 m, rRMSE = 39.80%). This model had similarities with other airborne LIDAR-based models where RH metrics were used as CBH predictors (Andersen et al., 2005; Engelstad et al., 2019; Stefanidou et al., 2020), although it obtained lower accuracy results compared to stand-level vegetation studies using ALS data (Hevia et al., 2016; Marino et al., 2022). Our results might also be influenced by the slope of the plots, ranging from 2.19 to 44.37% (calculated from the ALS), in that steep slopes have been shown to negatively affect GEDI ground elevation estimations (Adam et al., 2020), directly influencing RH metrics (Huettermann et al., 2022). However, in this study we did not filter out the steep slopes because we wanted to keep enough samples to be able to develop the CBH model.

The discontinuity of the spaceborne GEDI and its observation fringe of $\pm$51.6° N meant that the GEDI footprints had to be spatially extrapolated. This increased the uncertainty of the final maps and could be overcome in the future, once complete LiDAR coverage for the entire world is achieved (Hancock et al., 2021). Although GEDI did not provide wall-to-wall coverage, at 1-km resolution almost all the cells were sampled. If the length of the mission is extended, this coverage is expected to increase. We performed spatial extrapolation assuming the similarity of neighbouring forested areas belonging to the same forest fuel type and the dissimilarity of those belonging to different forest fuel types. This method was different from those used to generate other wall-to-wall maps on canopy fuel parameters covering large geographical areas (Lang et al., 2022; Potapov et al., 2021; Simard et al., 2011; Tang et al., 2019), which apply regression models to all the pixels or average the value of GEDI metrics, regardless of the fuel type polygon to which they belong or their proximity to other GEDI footprints.

The random forest regression models were strongly influenced by the training data, which should be representative of all the variability of European forest canopy fuel parameters, so as to be able to accurately predict fuel parameters at a European level. We tried to fulfil this requirement by choosing random samples stratified by fuel type on the basis of the distribution of the forest canopy parameter values. Moreover, low-quality training data that is not spatially or temporally coincident can reduce the accuracy of the models. This is why we decided to apply spatial (European forest only) and temporal (spring-summer GEDI-observed years) filters to the variables used in the models for spatial extrapolation. The random forest estimation models added uncertainty because they rely on external variables with their own limitations.

The Hm random forest regression model (r = 0.72–0.82, Bias = −0.18-0.29 m, RMSE = 3.63–4.18 m, rRMSE = 28.43–30.66%) obtained high levels of accuracy, similar for all fuel types. These results are not strictly comparable to other similar research studies. For instance, Potapov et al. (2021) obtained higher RMSE when extrapolating GEDI RH95 using a similar methodology to the random forest regression models applied here. We also obtained better results for the regression models compared to other global forest Hm maps that apply random forest regression models to RH100 from GLAS (Simard et al., 2011). Differences can also be found if we compare our results to the global map produced by Lang et al. (2022, RMSE = 2.7 m), which processes the GEDI L1B product and applies deep learning. These differences could be due to the fact that they used the L1B GEDI product instead of the L2A and L2B or perhaps because we used different machine and deep-learning approaches, as deep learning could be more suitable for highly complex parameter estimations in highly heterogeneous environments. The most important variable for our random forest regression was GEDI top of the canopy height (based on GEDI RH98) because of its similarity to the RH80 metric (RMSE = 10 m for Europe at 1 km

resolution). Another important variable was slope because of its influence on the correct detection of ground elevation by the full-waveform LiDAR. In line with other previous studies (Li et al., 2020; Nandy et al., 2021), SAR data was also important for estimating forest Hm, due to its relation with forest structure. Biophysical variables, temperature and precipitation were also important, as in previous papers (Simard et al., 2011; Tao et al., 2016; Zhang et al., 2016). Multispectral imagery and vegetation indices, used globally by Potapov et al. (2021), seemed less important than the other variables.

The CC random forest regression models (r = 0.82–0.91, Bias = 0, RMSE = 0.07–0.09, rRMSE = 10.65–14.42%) obtained similar accuracies for all fuel types, as obtained previously in local studies (Kacic et al., 2021). The final CC values were similar to other global CC products (DiMiceli et al., 2015), but showed differences compared to high-resolution European CC maps (European Environment Agency, 2020), as might be expected due to the lower spatial resolution of our product. One of the best explanatory variables for our model was top of the canopy height, given the relationship between tree height and crown size. HV SAR backscatter was also a very important variable as it is related to fuel moisture and biomass (Ningthoujam et al., 2016). Contrary to Tang et al. (2019), our models did not show strong differences by fuel types, which might be a result of only selecting GEDI footprints for the spring and summer, when both evergreen and deciduous forests have leaves.

As regards the accuracy of the CBH random forest regression model, similar results were obtained for all fuel types (r = 0.62–0.75, Bias = 0.01–0.02 m, RMSE = 0.60–0.74 m, rRMSE = 19.16–22.93%). Most of the previous works used ALS to model CBH from field data without applying any spatial extrapolation to areas not covered by LiDAR (Andersen et al., 2005; Chamberlain et al., 2021; Hall et al., 2005; Stefanidou et al., 2020). However, Erdody and Moskal (2010) generated continuous CBH maps using regression models that combined LiDAR and optical remote sensing imagery, obtaining better results (r = 0.84, RMSE = 0.68). This could be due to the smaller area studied in their research and the higher resolution of the data used. It is important to highlight that all previous studies were conducted at a local scale, while we made continental estimations based on a model trained at a regional scale (four Spanish provinces). The CBH models did not perform as well as the Hm and CC models because of the difficulty in predicting CBH with optical and SAR imagery and biophysical variables. Top of the canopy height was the most important variable when it came to extrapolating the model to Europe. This variable has a significantly positively correlation with CBH (Cameron et al., 2022; Martín-García et al., 2021; Meredieu and Caraglio, 2002) and can be used indirectly to obtain the structure of forested environments due to the relationship between tree height and forest structure.

The final European maps on Hm, CC and CBH provided information about the spatial distribution of these canopy fuel parameters over the continent. The Hm results obtained in our research improve on earlier studies based on national forest inventories (Moreno et al., 2017; Pucher et al., 2022) in that we provide spatial Hm information for the whole of Europe. For forest CC, the minimum (15%), maximum (98%) and mean (52%) CC values obtained in our research were in line with those obtained previously for European forest (Tang et al., 2019). For the forest CBH, the few incoherent pixels where forest Hm was lower than CBH were the result of the uncertainties of the models. European maps on forest Hm and CBH could be used to calculate canopy length by subtracting CBH from Hm for each pixel, which could be useful in forest management and fire behaviour modelling. However, these results should be taken cautiously, particularly in areas where CBH reference data were not available.

The uncertainty maps took into account error propagation in the different steps followed to obtain the final maps on Hm, CC and CBH. Higher uncertainty was frequently observed in mountainous areas, probably as a result of GEDI's limited ability to obtain correct metrics on steep slopes (Adam et al., 2020) or in areas with a mixed pattern of fuel types, probably because they make spatial extrapolation of fuel parameters more difficult. Higher relative errors are expected in low forests, whereas lower relative errors occur in taller forests. The use of different sources of information led to increase pixel-level uncertainty. Nevertheless, combining the final maps on canopy parameters with their uncertainty maps will provide a more realistic view of canopy fuel characterisation across Europe.

The final wall-to-wall European maps on canopy fuel parameters are an essential tool for describing, understanding and modelling forest crown fires in this continent, providing useful information for deriving fire severity, behaviour, propagation and emissions. They can also be used as input in canopy fire spread models (Cruz and Alexander, 2013), and fire-caused emissions and biomass consumption models, such as Consume (Prichard et al., 2006, 2020). The resulting maps could be used to further characterise European forests and their distribution. The parameters analysed here could be combined with European-scale surface fuel parameters –such as those analysed in Aragoneses et al. (2023)– as input in fire spread models such as FlamMap (Finney, 2006) or FARSITE (Finney, 2004) to provide a more comprehensive assessment of fire risk conditions. Furthermore, our results are easily updatable if new spaceborne LiDAR data becomes available, which would allow for efficient follow-up fire risk assessments over time.

Overall, spatially explicit continental-scale information on canopy fuel parameters is essential for reducing the risk of forest fires and their impacts (Chuvieco et al., 2023), and in particular the highly destructive extreme events that are commonly associated with crown fires. A holistic pan-European overview of fire risk would improve decision-making on fire prevention strategies, a key factor for building resilient ecosystems and communities in a rapidly changing world in which, as a general trend, climate change is constantly increasing the risk of forest fires. We used GEDI to develop methods to estimate critical canopy fuel parameters (Hm, CC and CBH) that can help us understand more about forest fire risk. This is the first attempt to generate fire risk prevention-oriented maps on canopy fuel parameters at a European level, and is particularly important in terms of CBH, for which no large-scale maps had been developed before, although the difficulty associated with estimating CBH and the fact that this parameter could only be calibrated for 4 areas in Spain must also be taken into account. To summarize, this research makes an important contribution to fire risk prevention in Europe, as it spatially describes some of the most relevant characteristics of forest canopy fuels.

Future research should focus on applying similar methods to further increase CBH reference estimations or to estimate other relevant canopy fuel parameters, such as canopy fuel load (Leite et al., 2022) or canopy bulk density (García et al., 2012). The methods developed here could be applied in Europe and beyond, for example in the calibration of different models for different areas and regions of the world, which hopefully would obtain similar high-quality results to those obtained here.

5. Conclusions

This work has demonstrated the potential of the spaceborne LiDAR Global Ecosystem Dynamics Investigation (GEDI) and other geospatial data and remote sensing images to estimate various forest canopy fuel parameters that affect crown fire behaviour and impacts, namely forest canopy height (Hm), canopy cover (CC) and canopy base height (CBH), at the European continental scale. Strong correlations were found between GEDI RH80 and ALS Hm, GEDI CC and ALS CC, and between field-based and GEDI-derived CBH. GEDI's sampling acquisition characteristics made it necessary to spatially extrapolate the estimated canopy fuel parameters from GEDI to generate continuous Europe-wide maps. This involved the spatial interpolation of GEDI footprints and the development of regression models by fuel types using auxiliary geospatial data. The uncertainty maps provided for the different parameters took into consideration the propagation of individual errors for each step in the methodology.

The resulting wall-to-wall maps on canopy fuel parameters in Europe, which are publicly available (Aragoneses et al., 2024) https://doi.org/10.21950/KTALA8, constitute the first attempt to create continental maps on forest canopy fuel parameters calibrated at a European scale that aim to reduce the risk of forest fires. The achievement in terms of CBH is especially important in that it had never been represented on such large-scale maps before. Our results, together with the European fuel type map and surface fuel models (Aragoneses et al., 2023), could be useful for reducing the fire risk in Europe as they offer a holistic overview of the distribution of canopy fuel parameters across the continent, as well as serving as pixel-level input layers in fire behaviour or emissions models. In order to ensure a comprehensive European-wide approach to fire risk that can help build fire-resilient ecosystems and communities, future research must focus on applying similar methods to estimate other relevant canopy fuel parameters at a European scale, such as fuel load or canopy bulk density.

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CRediT authorship contribution statement

Elena Aragoneses: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Mariano García:** Conceptualization, Methodology, Supervision, Writing – review & editing. **Paloma Ruiz-Benito:** Data curation, Methodology, Resources, Writing – review & editing. **Emilio Chuvieco:** Conceptualization, Funding acquisition, Methodology, Project administration, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The resulting European maps on canopy height, canopy cover and canopy base height, together with their uncertainties (circa 2020; 1 km spatial resolution) are publicly available at https://doi.org/10.21950/KTALA8 from e-cienciaDatos Aragoneses et al. (2024).

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rse.2024.114005.

References

Adam, M., Urbazaev, M., Dubois, C., Schmullius, C., 2020. Accuracy assessment of gedi terrain elevation and canopy height estimates in European temperate forests: influence of environmental and acquisition parameters. Remote Sens. 12, 1–28. https://doi.org/10.3390/rs12233948.

Alberdi, I., Sandoval, V., Condés, S., Cañellas, I., Vallejo, R., 2016. El Inventario Forestal Nacional español, una herramienta para el conocimiento, la gestión y la conservación de los ecosistemas forestales arbolados. Ecosistemas 25, 88–97. https://doi.org/10.7818/ECOS.2016.25-3.10.

Albini, F., 1976. Estimating Wildfire Behavior and Effects. UT, USA.

Almeida-Filho, R., Shimabukuro, Y.E., Rosenqvist, A., Sánchez, G.A., 2009. Using dual-polarized ALOS PALSAR data for detecting new fronts of deforestation in the Brazilian Amazônia. Int. J. Remote Sens. 30, 3735–3743. https://doi.org/10.1080/01431160902777175.

Alonso-Rego, C., Arellano-Pérez, S., Guerra-Hernández, J., Molina-Valero, J.A., Martínez-Calvo, A., Pérez-Cruzado, C., Castedo-Dorado, F., González-Ferreiro, E., Álvarez-González, J.G., Ruiz-González, A.D., 2021. Estimating stand and fire-related surface and canopy fuel variables in pine stands using low-density airborne and single-scan terrestrial laser scanning data. Remote Sens. 13, 5170. https://doi.org/10.3390/RS13245170.

Andersen, H.E., McGaughey, R.J., Reutebuch, S.E., 2005. Estimating forest canopy fuel parameters using LIDAR data. Remote Sens. Environ. 94, 441–449. https://doi.org/10.1016/j.rse.2004.10.013.

Anderson, H., 1982. Aids to Determining Fuel Models for Estimating Fire Behavior. Washington, DC, USA.

Aragoneses, E., Chuvieco, E., 2021. Generation and mapping of fuel types for fire risk assessment. Fire 4 (3), 1–26. https://doi.org/10.3390/FIRE4030059.

Aragoneses, E., García, M., Chuvieco, E., 2022. FirEUrisk_Europe_fuel_map: European fuel map at 1 km resolution, e-cienciaDatos doi:doi:10.21950/YABYCN.

Aragoneses, E., García, M., Salis, M., Ribeiro, L.M., Chuvieco, E., 2023. Classification and mapping of European fuels using a hierarchical, multipurpose fuel classification system. Earth Syst. Sci. Data 15, 1287–1315. https://doi.org/10.5194/ESSD-15-1287-2023.

Aragoneses, E., García, M., Chuvieco, E., 2024. FirEUrisk_canopy_fuel_parameters: canopy height, canopy cover and canopy base height. e-cienciaDatos. https://doi.org/10.21950/KTALA8.

Arroyo, L.A., Pascual, C., Manzanera, J.A., 2008. Fire models and methods to map fuel types: the role of remote sensing. For. Ecol. Manag. 256, 1239–1252. https://doi.org/10.1016/j.foreco.2008.06.048.

Artés, T., Oom, D., de Rigo, D., Durrant, T.H., Maianti, P., Libertà, G., San-Miguel-Ayanz, J., 2019. A global wildfire dataset for the analysis of fire regimes and fire behaviour. Sci. Data 6, 1–11. https://doi.org/10.1038/s41597-019-0312-2.

Arumäe, T., Lang, M., 2017. Estimation of canopy cover in dense mixed-species forests using airborne lidar data. Eur. J. Remote Sens. 51, 132–141. https://doi.org/10.1080/22797254.2017.1411169.

Banco de Datos de la Naturaleza, 2022. Mapa Forestal de España. Máxima actualidad, Ministerio para la Transición Ecológica y el Reto Demográfico.

Breiman, L., Friedman, J.H., Olshen, R.A., Stone, C.J., 1984. Classification and Regression Trees, Classification and Regression Trees. Routledge, New York. https://doi.org/10.1201/9781315139470.

Burnham, K.P., Anderson, D.R., 2002. Model selection and multimodel inference. In: Model Selection and Multimodel Inference. Springer, New York. https://doi.org/10.1007/B97636.

Cameron, H.A., Schroeder, D., Beverly, J.L., 2022. Predicting black spruce fuel characteristics with airborne laser scanning (ALS). Int. J. Wildland Fire 31, 124–135. https://doi.org/10.1071/WF21004.

Cardoso Castro Rego, F.M., Moreno Rodríguez, J.M., Vallejo Calzada, V.R., Xanthopoulos, G., 2018. Forest fires. In: Sparking policies in the EU. European Commission. https://doi.org/10.2777/248004.

Cavanaugh, J.E., Neath, A.A., 2019. The Akaike information criterion: background, derivation, properties, application, interpretation, and refinements. Wiley Interdiscip. Rev. Comput. Stat. 11, 1460. https://doi.org/10.1002/WICS.1460.

Centro de Descargas del CNIG (IGN), 2023. Instituto Geográfico Nacional (IGN).

Chamberlain, C.P., Sánchez Meador, A.J., Thode, A.E., 2021. Airborne lidar provides reliable estimates of canopy base height and canopy bulk density in southwestern ponderosa pine forests. For. Ecol. Manag. 481 https://doi.org/10.1016/j.foreco.2020.118695.

Chen, Q., 2010. Retrieving vegetation height of forests and woodlands over mountainous areas in the Pacific coast region using satellite laser altimetry. Remote Sens. Environ. 114, 1610–1627. https://doi.org/10.1016/J.RSE.2010.02.016.

Chuvieco, E., Aguado, I., Salas, J., García, M., Yebra, M., Oliva, P., 2020. Satellite remote sensing contributions to wildland fire science and management. Curr. For. Rep. 6, 81–96. https://doi.org/10.1007/s40725-020-00116-5.

Chuvieco, E., Yebra, M., Martino, S., Thonicke, K., Gómez-Giménez, M., San-Miguel, J., Oom, D., Velea, R., Mouillot, F., Molina, J.R., Miranda, A.I., Lopes, D., Salis, M., Bugaric, M., Sofiev, M., Kadantsev, E., Gitas, I.Z., Stavrakoudis, D., Eftychidis, G., Bar-Massada, A., Neidermeier, A., Pampanoni, V., Pettinari, M.L., Arrogante-Funes, F., Ochoa, C., Moreira, B., Viegas, D., 2023. Towards an integrated approach to wildfire risk assessment: when, where, what and how may the landscapes burn. Fire 6. https://doi.org/10.3390/FIRE6050215 page 215 6, 215.

Countryman, C., 1972. The Fire Environment Concept. USDA Forest Service, Pacific Southwest Range and Experiment Station, Berkeley, California, USA.

Cruz, M.G., Alexander, M.E., 2013. Uncertainty associated with model predictions of surface and crown fire rates of spread. Environ. Model Softw. 47, 16–28. https://doi.org/10.1016/J.ENVSOFT.2013.04.004.

DiMiceli, C., Carroll, M., Sohlberg, R., Kim, D., Kelly, M., Townshend, J., 2015. MOD44B MODIS/Terra Vegetation Continuous Fields Yearly L3 Global 250m SIN Grid V006 [Data Set]. NASA EOSDIS L. Process. DAAC. https://doi.org/10.5067/MODIS/MOD44B.006.

Dorado-Roda, I., Pascual, A., Godinho, S., Silva, C.A., Botequim, B., Rodríguez-Gonzálvez, P., González-Ferreiro, E., Guerra-Hernández, J., 2021. Assessing the accuracy of gedi data for canopy height and aboveground biomass estimates in mediterranean forests. Remote Sens. 13 https://doi.org/10.3390/rs13122279.

Duane, A., Castellnou, M., Brotons, L., 2021. Towards a comprehensive look at global drivers of novel extreme wildfire events. Clim. Chang. 165, 1–21. https://doi.org/10.1007/S10584-021-03066-4.

Dubayah, R., Blair, J.B., Goetz, S., Fatoyinbo, L., Hansen, M., Healey, S., Hofton, M., Hurtt, G., Kellner, J., Luthcke, S., Armston, J., Tang, H., Duncanson, L., Hancock, S., Jantz, P., Marselis, S., Patterson, P.L., Qi, W., Silva, C., 2020. The global ecosystem dynamics investigation: high-resolution laser ranging of the Earth's forests and topography. Sci. Remote Sens. 1, 100002 https://doi.org/10.1016/j.srs.2020.100002.

Engelstad, P.S., Falkowski, M., Wolter, P., Poznanovic, A., Johnson, P., 2019. Estimating canopy fuel attributes from low-density lidar. Fire 2, 1–19. https://doi.org/10.3390/fire2030038.

Erdody, T.L., Moskal, L.M., 2010. Fusion of LiDAR and imagery for estimating forest canopy fuels. Remote Sens. Environ. 114, 725–737. https://doi.org/10.1016/j.rse.2009.11.002.

European Environment Agency, 2016. Biogeographical regions.

European Environment Agency, 2020. High Resolution Layer: Tree Cover Density (TCD) 2018, Copernicus Land Monitoring Service.

European Forest Fire Information System, 2017. European Fuel Map, based on JRC Contract Number 384347 on the "Development of a European Fuel Map.".

Ferraz, A., Bretar, F., Jacquemoud, S., Gonçalves, G., 2009. The Role of Lidar Systems in Fuel Mapping. Instituto de Engenharia de Sistemas e Computadores de Coimbra, Coimbra, Portugal.

Fick, S.E., Hijmans, R.J., 2017. WorldClim 2: new 1km spatial resolution climate surfaces for global land areas. Int. J. Climatol. 37, 4302–4315. https://doi.org/10.1002/joc.5086.

Finney, M.A., 2004. FARSITE: Fire Area Simulator – Model Development and Evaluation.

Finney, M.A., 2006. An overview of FlamMap Fire modeling capabilities. In: Andrews, P. L., Butler, B.W. (Eds.), Fuels Management-How to Measure Success. Conference Proceedings, 28–30 March 2006, Portland, OR, Proceedings RMRS-P-41, Fort Collins, CO, U.S. Department of Agriculture, Forest Service, Rocky Mountain Research Station, pp. 213–220.

Flood, N., 2013. Seasonal composite Landsat TM/ETM+ images using the medoid (a multi-dimensional median). Remote Sens. 5, 6481–6500. https://doi.org/10.3390/RS5126481.

Fu, L., Zhang, H., Sharma, R.P., Pang, L., Wang, G., 2017. A generalized nonlinear mixed-effects height to crown base model for Mongolian oak in Northeast China. For. Ecol. Manag. 384, 34–43. https://doi.org/10.1016/J.FORECO.2016.09.012.

Gajardo, J., García, M., Riaño, D., 2014. Applications of airborne laser scanning in Forest fuel assessment and fire prevention. In: Maltamo, M., Næsset, E., Vauhkonen, J. (Eds.), Forestry Applications of Airborne Laser Scanning. Managing Forest Ecosystems. Springer, Dordrecht, pp. 439–462. https://doi.org/10.1007/978-94-017-8663-8_22.

Gale, M.G., Cary, G.J., Van Dijk, A.I.J.M., Yebra, M., 2021. Forest fire fuel through the lens of remote sensing: review of approaches, challenges and future directions in the remote sensing of biotic determinants of fire behaviour. Remote Sens. Environ. 255, 112282 https://doi.org/10.1016/j.rse.2020.112282.

Gao, B.C., 1996. NDWI—A normalized difference water index for remote sensing of vegetation liquid water from space. Remote Sens. Environ. 58, 257–266. https://doi.org/10.1016/S0034-4257(96)00067-3.

García, M., Danson, F.M., Riaño, D., Chuvieco, E., Ramirez, F.A., Bandugula, V., 2011. Terrestrial laser scanning to estimate plot-level forest canopy fuel properties. Int. J. Appl. Earth Obs. Geoinf. 13, 636–645. https://doi.org/10.1016/J.JAG.2011.03.006.

García, M., Popescu, S., Riaño, D., Zhao, K., Neuenschwander, A., Agca, M., Chuvieco, E., 2012. Characterization of canopy fuels using ICESat/GLAS data. Remote Sens. Environ. 123, 81–89. https://doi.org/10.1016/j.rse.2012.03.018.

García, M., Saatchi, S., Casas, A., Koltunov, A., Ustin, S.L., Ramirez, C., Balzter, H., 2017. Extrapolating forest canopy fuel properties in the California rim fire by combining airborne LiDAR and landsat OLI data. Remote Sens. 9, 1–18. https://doi.org/10.3390/rs9040394.

García, M., Saatchi, S., Ustin, S., Balzter, H., 2018. Modelling forest canopy height by integrating airborne LiDAR samples with satellite Radar and multispectral imagery. Int. J. Appl. Earth Obs. Geoinf. 66, 159–173. https://doi.org/10.1016/J.JAG.2017.11.017.

GEOportal.NRW, 2023. GEOportal.NRW.

German Aerospace Center (DLR), 2018. TanDEM-X - Digital Elevation Model (DEM) - Global, 90m. https://doi.org/10.15489/ju28hc7pui09.

Gitelson, A.A., Kaufman, Y.J., Merzlyak, M.N., 1996. Use of a green channel in remote sensing of global vegetation from EOS-MODIS. Remote Sens. Environ. 58, 289–298. https://doi.org/10.1016/s0034-4257(96)00072-7.

Hall, S.A., Burke, I.C., Box, D.O., Kaufmann, M.R., Stoker, J.M., 2005. Estimating stand structure using discrete-return lidar: an example from low density, fire prone ponderosa pine forests. For. Ecol. Manag. 208, 189–209. https://doi.org/10.1016/J.FORECO.2004.12.001.

Hancock, S., Armston, J., Hofton, M., Sun, X., Tang, H., Duncanson, L.I., Kellner, J.R., Dubayah, R., 2019. The GEDI simulator: a large-footprint waveform lidar simulator for calibration and validation of spaceborne missions. Earth Sp. Sci. 6, 294–310. https://doi.org/10.1029/2018EA000506.

Hancock, S., McGrath, C., Lowe, C., Davenport, I., Woodhouse, I., 2021. Requirements for a global lidar system: spaceborne lidar with wall-to-wall coverage. R. Soc. Open Sci. 8 https://doi.org/10.1098/RSOS.211166.

Hansen, M.C., Potapov, P.V., Moore, R., Hancher, M., Turubanova, S.A., Tyukavina, A., Thau, D., Stehman, S.V., Goetz, S.J., Loveland, T.R., Kommareddy, A., Egorov, A., Chini, L., Justice, C.O., Townshend, J.R.G., 2013. High-resolution global maps of 21st-century forest cover change. Science (80-.). 342, 850–853. https://doi.org/10.1126/SCIENCE.1244693.

Harrell, F.E., Lee, K.L., Califf, R.M., Pryor, D.B., Rosati, R.A., 1984. Regression modelling strategies for improved prognostic prediction. Stat. Med. 3, 143–152. https://doi.org/10.1002/SIM.4780030207.

Healey, S.P., Yang, Z., Gorelick, N., Ilyushchenko, S., 2020. Highly local model calibration with a new GEDI LiDAR asset on Google earth engine reduces Landsat Forest height signal saturation. Remote Sens. 12, 2840. https://doi.org/10.3390/RS12172840.

Hevia, A., Álvarez-González, J.G., Ruiz-Fernández, E., Prendes, C., Ruiz-González, A.D., Majada, J., González-Ferreiro, E., 2016. Modelling canopy fuel and forest stand variables and characterizing the influence of thinning in the stand structure using airborne LiDAR. Rev. Teledetección Rev. la Asoc. Española Teledetección 45, 41–55.

Hijmans, R.J., Cameron, S.E., Parra, J.L., Jones, P.G., Jarvis, A., 2005. Very high resolution interpolated climate surfaces for global land areas. Int. J. Climatol. 25, 1965–1978. https://doi.org/10.1002/JOC.1276.

Huete, A.R., Liu, H.Q., Batchily, K., Van Leeuwen, W., 1997. A comparison of vegetation indices over a global set of TM images for EOS-MODIS. Remote Sens. Environ. 59, 440–451. https://doi.org/10.1016/S0034-4257(96)00112-5.

Huettermann, S., Jones, S., Soto-Berelov, M., Hislop, S., 2022. Intercomparison of real and simulated GEDI observations across Sclerophyll forests. Remote Sens. 2022 (14), 2096. https://doi.org/10.3390/RS14092096.

IPCC, 2022. Climate change 2022: Impacts, adaptation, and vulnerability. In: Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge University Press. In Press.

James, G., Witten, D., Hastie, T., Tibshirani, R., 2013. An Introduction to Statistical Learning with Applications in R.

Jones, M.W., Abatzoglou, J.T., Veraverbeke, S., Andela, N., Lasslop, G., Forkel, M., Smith, A.J.P., Burton, C., Betts, R.A., van der Werf, G.R., Sitch, S., Canadell, J.G., Santín, C., Kolden, C., Doerr, S.H., Le Quéré, C., 2022. Global and regional trends and drivers of fire under climate change. Rev. Geophys. 60 https://doi.org/10.1029/2020RG000726.

Kacic, P., Hirner, A., Da Ponte, E., 2021. Fusing Sentinel-1 and -2 to model GEDI-derived vegetation structure characteristics in GEE for the Paraguayan Chaco. Remote Sens. 2021 (13), 5105. https://doi.org/10.3390/RS13245105.

Keane, R.E., Reinhardt, E.D., Scott, J., Gray, K., Reardon, J., 2005. Estimating forest canopy bulk density using six indirect methods. Can. J. For. Res. 35, 724–739. https://doi.org/10.1139/X04-213.

Kelly, L.T., Giljohann, K.M., Duane, A., Aquilué, N., Archibald, S., Batllori, E., Bennett, A. F., Buckland, S.T., Canelles, Q., Clarke, M.F., Fortin, M.J., Hermoso, V., Herrando, S., Keane, R.E., Lake, F.K., McCarthy, M.A., Morán-Ordóñez, A., Parr, C.L., Pausas, J.G., Penman, T.D., Regos, A., Rumpff, L., Santos, J.L., Smith, A.L., Syphard, A.D., Tingley, M.W., Brotons, L., 2020. Fire and biodiversity in the Anthropocene. Science (80-.). 370. https://doi.org/10.1126/science.abb0355.

Köble, R., Seufert, G., 2001. Novel maps for forest tree species in Europe. In: Proc. 8th Eur. Symp. Physico-Chemical Behav. Air Pollut. A Chang. Atmos.

Lang, N., Jetz, W., Schindler, K., Wegner, J.D., 2022. A High-Resolution Canopy Height Model of the Earth. arXiv arXiv:2204.

Laserscandaten, 2023. Geobroker - Der Internetshop der LGB.

Lefsky, M.A., 2010. A global forest canopy height map from the moderate resolution imaging Spectroradiometer and the geoscience laser altimeter system. Geophys. Res. Lett. 37 https://doi.org/10.1029/2010GL043622.

Leite, R.V., Silva, C.A., Broadbent, E.N., do Amaral, C.H., Liesenberg, V., de Almeida, D. R.A., Mohan, M., Godinho, S., Cardil, A., Hamamura, C., de Faria, B.L., Brancalion, P.H.S., Hirsch, A., Marcatti, G.E., Dalla Corte, A.P., Zambrano, A.M.A., da Costa, M.B.T., Matricardi, E.A.T., da Silva, A.L., Goya, L.R.R.Y., Valbuena, R., de Mendonça, B.A.F., Silva Junior, C.H.L., Aragão, L.E.O.C., García, M., Liang, J., Merrick, T., Hudak, A.T., Xiao, J., Hancock, S., Duncason, L., Ferreira, M.P., Valle, D., Saatchi, S., Klauberg, C., 2022. Large scale multi-layer fuel load characterization in tropical savanna using GEDI spaceborne lidar data. Remote Sens. Environ. 268 https://doi.org/10.1016/J.RSE.2021.112764.

Li, F., Lawrence, D.M., Bond-Lamberty, B., 2018. Human impacts on 20th century fire dynamics and implications for global carbon and water trajectories. Glob. Planet. Chang. 162, 18–27. https://doi.org/10.1016/J.GLOPLACHA.2018.01.002.

Li, W., Niu, Z., Shang, R., Qin, Y., Wang, L., Chen, H., 2020. High-resolution mapping of forest canopy height using machine learning by coupling ICESat-2 LiDAR with Sentinel-1, Sentinel-2 and Landsat-8 data. Int. J. Appl. Earth Obs. Geoinf. 92, 102163 https://doi.org/10.1016/j.jag.2020.102163.

Lidar GIS viewer, 2023. Agencija rs za okolje.

Liu, A., Cheng, X., Chen, Z., 2021. Performance evaluation of GEDI and ICESat-2 laser altimeter data for terrain and canopy height retrievals. Remote Sens. Environ. 264, 112571 https://doi.org/10.1016/j.rse.2021.112571.

Lizundia-Loiola, J., Franquesa, M., Khairoun, A., Chuvieco, E., 2022. Global burned area mapping from Sentinel-3 synergy and VIIRS active fires. Remote Sens. Environ. 282, 113298 https://doi.org/10.1016/J.RSE.2022.113298.

Lloret, F., Zedler, P.H., 2009. The effect of Forest fire on vegetation. In: Cerda, A. (Ed.), Fire Effects on Soils and Restoration Strategies. CRC Press, Boca Raton, pp. 273–312. https://doi.org/10.1201/9781439843338-13.

Luo, L., Zhai, Q., Su, Y., Ma, Q., Kelly, M., Guo, Q., 2018. Simple method for direct crown base height estimation of individual conifer trees using airborne LiDAR data. Opt. Express 26, A562. https://doi.org/10.1364/oe.26.00a562.

Marino, E., Tomé, J.L., Hernando, C., Guijarro, M., Madrigal, J., 2022. Transferability of airborne LiDAR data for canopy fuel mapping: effect of pulse density and model formulation. Fire 5, 126. https://doi.org/10.3390/fire5050126.

Martín-García, S., Banelovic, I., Jurevic, L., Lizarralde, I., Indir, K., Ponce, R.A., 2021. Height to Crown Base Modelling for the Main Tree Species in Even-Aged Pedunculate Oak Forest: A Case Study from Central Croatia. SEFOR South-east Eur. For, 21, pp. 1–11. https://doi.org/10.15177/seefor.21-04.

Meredieu, C., Caraglio, Y., 2002. External indicators of living branches with missing rings within a tree crown of Corsican pine. For. An Int. J. For. Res. 75, 569–578. https://doi.org/10.1093/FORESTRY/75.5.569.

Miller, J.D., Thode, A.E., 2007. Quantifying burn severity in a heterogeneous landscape with a relative version of the delta normalized burn ratio (dNBR). Remote Sens. Environ. 109, 66–80. https://doi.org/10.1016/J.RSE.2006.12.006.

Moreno, A., Neumann, M., Hasenauer, H., 2017. Forest structures across Europe. Geosci. Data J. 4, 17–28. https://doi.org/10.1002/GDJ3.45.

Morin, D., Planells, M., Guyon, D., Villard, L., Mermoz, S., Bouvet, A., Thevenon, H., Dejoux, J.F., Le Toan, T., Dedieu, G., 2019. Estimation and mapping of Forest structure parameters from open access satellite images: development of a generic method with a study case on coniferous plantation. Remote Sens. 2019 (11), 1275. https://doi.org/10.3390/RS11111275.

Nandy, S., Srinet, R., Padalia, H., 2021. Mapping forest height and aboveground biomass by integrating ICESat-2, sentinel-1 and Sentinel-2 data using random Forest algorithm in Northwest Himalayan foothills of India. Geophys. Res. Lett. 48, 1–10. https://doi.org/10.1029/2021GL093799.

Ningthoujam, R.K., Tansey, K., Balzter, H., Morrison, K., Johnson, S.C.M., Gerard, F., George, C., Burbidge, G., Doody, S., Veck, N., Llewellyn, G.M., Blythe, T., 2016. Mapping forest cover and forest cover change with airborne S-band radar. Remote Sens. 8, 577. https://doi.org/10.3390/RS8070577.

Oliveira, S., Gonçalves, A., Benali, A., Sá, A., Zêzere, J.L., Pereira, J.M., 2020. Assessing risk and prioritizing safety interventions in human settlements affected by large wildfires. Forests 11, 859. https://doi.org/10.3390/F11080859.

Openko, I., Tykhenko, R., Shevchenko, O., Tsvyakh, O., Stepchuk, Y., 2023. Mathematical modeling of economic losses caused by forest fire in Ukraine. In: Handbook of Research on Improving the Natural and Ecological Conditions of the Polesie Zone. IGI Global, pp. 372–383. https://doi.org/10.4018/978-1-6684-8248-3.CH023.

Pinheiro, J., Bates, D., R Core Team, 2023. nnlme: Linear and Nonlinear Mixed Effects Models, R package version 3.1-162. https://CRAN.R-project.org/package=nlme.

Pirotti, F., 2011. Analysis of full-waveform LiDAR data for forestry applications: a review of investigations and methods. IForest 4, 100–106. https://doi.org/10.3832/ifor0562-004.

Potapov, P., Li, X., Hernandez-Serna, A., Tyukavina, A., Hansen, M.C., Kommareddy, A., Pickens, A., Turubanova, S., Tang, H., Silva, C.E., Armston, J., Dubayah, R., Blair, J. B., Hofton, M., 2021. Mapping global forest canopy height through integration of GEDI and Landsat data. Remote Sens. Environ. 253, 112165 https://doi.org/10.1016/j.rse.2020.112165.

Prichard, S.J., Ottmar, R.D., Anderson, G.K., 2006. Consume 3.0 User's Guide. Seattle, Washington.

Prichard, S.J., O'Neill, S.M., Eagle, P., Andreu, A.G., Drye, B., Dubowy, J., Urbanski, S., Strand, T.M., Prichard, S.J., O'Neill, S.M., Eagle, P., Andreu, A.G., Drye, B., Dubowy, J., Urbanski, S., Strand, T.M., 2020. Wildland fire emission factors in North America: synthesis of existing data, measurement needs and management applications. Int. J. Wildland Fire 29, 132–147. https://doi.org/10.1071/WF19066.

Pucher, C., Neumann, M., Hasenauer, H., 2022. An improved forest structure data set for Europe. Remote Sens. 14, 395. https://doi.org/10.3390/RS14020395.

Quinn, G.P., Keough, M.J., 2002. Experimental Design and Data Analysis for Biologists. Cambridge Cambridge Univ. Press. https://doi.org/10.1017/CBO9780511806384.

Quiros, E., Polo, M.E., Fragoso-Campon, L., 2021. GEDI elevation accuracy assessment: a case study of Southwest Spain. IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. 14, 5285–5299. https://doi.org/10.1109/JSTARS.2021.3080711.

R Core Team, 2021. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria.

Rawat, D., Khanduri, V.P., Sati, S.P., Ahmad, T., Riyal, M., Mishra, P., 2020. Impact of forest fire on soil quality and resilience potential: a review. ADRRI J. Agric. Food Sci. 4, 59–81. https://doi.org/10.55058/ADRRIJAFS.V4I6(4).606.

Riaño, D., Meier, E., Allgöwer, B., Chuvieco, E., Ustin, S.L., 2003. Modeling airborne laser scanning data for the spatial generation of critical forest parameters in fire behavior modeling. Remote Sens. Environ. 86, 177–186. https://doi.org/10.1016/S0034-4257(03)00098-1.

Riaño, D., Chuvieco, E., Condés, S., González-Matesanz, J., Ustin, S.L., 2004. Generation of crown bulk density for Pinus sylvestris L. from lidar. Remote Sens. Environ. 92, 345–352. https://doi.org/10.1016/J.RSE.2003.12.014.

Rishmawi, K., Huang, C., Zhan, X., 2021. Monitoring key forest structure attributes across the conterminous United States by integrating gedi lidar measurements and viirs data. Remote Sens. 13, 1–23. https://doi.org/10.3390/rs13030442.

Rouse Jr., J.W., Haas, R.H., Schell, J.A., Deering, D.W., 1974. Monitoring vegetation systems in the great plains with erts. NASA Spec. Publ. 309–350.

Roy, D.P., Kashongwe, H.B., Armston, J., 2021. The impact of geolocation uncertainty on GEDI tropical forest canopy height estimation and change monitoring. Sci. Remote Sens. 4, 100024 https://doi.org/10.1016/J.SRS.2021.100024.

Ruiz-Benito, P., Vacchiano, G., Lines, E.R., Reyer, C.P.O., Ratcliffe, S., Morin, X., Hartig, F., Mäkelä, A., Yousefpour, R., Chaves, J.E., Palacios-Orueta, A., Benito-Garzón, M., Morales-Molino, C., Camarero, J.J., Jump, A.S., Kattge, J., Lehtonen, A., Ibrom, A., Owen, H.J.F., Zavala, M.A., 2020. Available and missing data to model impact of climate change on European forests. Ecol. Model. 416, 108870 https://doi.org/10.1016/J.ECOLMODEL.2019.108870.

Running, S., Zhao, M., 2019. MOD17A3HGF MODIS/Terra Net Primary Production Gap-Filled Yearly L4 Global 500 M SIN Grid V006 [Data Set]. NASA EOSDIS L. Process. DAAC. https://doi.org/10.5067/MODIS/MOD17A3HGF.006.

San-Miguel-Ayanz, J., Durrant, T., Boca, R., Maianti, P., Liberta`, G., Artes Vivancos, T., Jacome Felix Oom, D., Branco, A., De Rigo, D., Ferrari, D., Pfeiffer, H., Grecchi, R., Nuijten, D., Onida, M., Loffler, P., 2021a. Forest Fires in Europe, Middle East and North Africa 2020. In: Publ. Off. Eur. Union, Luxemb, pp. 1–174. https://doi.org/10.2760/216446.

San-Miguel-Ayanz, J., Durrant, T., Boca, R., Maianti, P., Libertá, G., Artés-Vivancos, T., Oom, D., Branco, A., de Rigo, D., Ferrari, D., Pfeiffer, H., Grecchi, R., Nuijten, D., Onida, M., 2021b. Advance EFFIS report on forest fires in Europe, Middle East and North Africa 2020. In: Publ. Off. Eur. Union, Luxemb. https://doi.org/10.2760/344684.

San-Miguel-Ayanz, J., Durrant, T., Boca, R., Maianti, P., Libertá, G., Artés-Vivancos, T., Oom, D., Branco, A., de Rigo, D., Ferrari, D., Pfeiffer, H., Grecchi, R., Nuijten, D., 2022. Advance Report on Wildfires in Europe, Middle East and North Africa 2021. Publ. Off. Eur. Union, Luxemb, pp. 1–37.

San-Miguel-Ayanz, J., Durrant, T., Boca, R., Maianti, P., LastNameLibertà, G., Oom, D., Branco, A., De Rigo, D., Ferrari, D., Roglia, E., Scionti, N., 2023. Advance report on forest fires in Europe, Middle East and North Africa 2022. In: Publ. Off. Eur. Union, Luxemb, pp. 1–54. https://doi.org/10.2760/091540.

Scott, J., Burgan, R., 2005. Standard fire behavior fuel models: A comprehensive set for use with Rothermel's surface fire spread model. In: US Department of Agriculture, Forest Service, Rocky Mountain Research Station. Rocky Mountain Research Station.

Sharma, R.P., Vacek, Z., Vacek, S., Podrázský, V., Jansa, V., 2017. Modelling individual tree height to crown base of Norway spruce (Picea abies (L.) karst.) and European beech (Fagus sylvatica L.). PLoS One 12. https://doi.org/10.1371/JOURNAL.PONE.0186394.

Shimada, M., Itoh, T., Motooka, T., Watanabe, M., Shiraishi, T., Thapa, R., Lucas, R., 2014. New global forest/non-forest maps from ALOS PALSAR data (2007–2010). Remote Sens. Environ. 155, 13–31. https://doi.org/10.1016/J.RSE.2014.04.014.

Silva, C.A., Saatchi, S., Garcia, M., Labriere, N., Klauberg, C., Ferraz, A., Meyer, V., Jeffery, K.J., Abernethy, K., White, L., Zhao, K., Lewis, S.L., Hudak, A.T., 2018. Comparison of small-and large-footprint lidar characterization of tropical Forest aboveground structure and biomass: a case study from Central Gabon. IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. 11, 3512–3526. https://doi.org/10.1109/JSTARS.2018.2816962.

Silva, C., Hamamura, C., Valbuena, R., Hancock, S., Cardil, A., Broadbent, E.N., Almeida, D.R.A., Silva Junior, C.H., Klauberg, C., 2020. rGEDI: NASA's Global Ecosystem Dynamics Investigation (GEDI) Data Visualization and Processing.

Silveira, E.M.O., Radeloff, V.C., Martinuzzi, S., Martinez Pastur, G.J., Bono, J., Politi, N., Lizarraga, L., Rivera, L.O., Ciuffoli, L., Rosas, Y.M., Olah, A.M., Gavier-Pizarro, G.I., Pidgeon, A.M., 2023. Nationwide native forest structure maps for Argentina based on forest inventory data, SAR Sentinel-1 and vegetation metrics from Sentinel-2 imagery. Remote Sens. Environ. 285, 113391 https://doi.org/10.1016/J.RSE.2022.113391.

Simard, M., Pinto, N., Fisher, J.B., Baccini, A., 2011. Mapping forest canopy height globally with spaceborne lidar. J. Geophys. Res. 116, 1–12. https://doi.org/10.1029/2011JG001708.

Smith, H.G., Sheridan, G.J., Lane, P.N.J., Nyman, P., Haydon, S., 2011. Wildfire effects on water quality in forest catchments: a review with implications for water supply. J. Hydrol. 396, 170–192. https://doi.org/10.1016/J.JHYDROL.2010.10.043.

Stefanidou, A., Gitas, I.Z., Korhonen, L., Stavrakoudis, D., Georgopoulos, N., 2020. LiDAR-based estimates of canopy base height for a dense uneven-aged structured forest. Remote Sens. 12, 1–20. https://doi.org/10.3390/rs12101565.

Takaku, J., Tadono, T., Tsutsui, K., 2014. Generation of hight resolution global DSM from ALOS PRISM. Int. Arch. Photogramm. Remote Sens. Spat. Inf. Sci. ISPRS Tech. Comm. IV Symp. XL–4, 243–248. https://doi.org/10.5194/isprsarchives-XL-4-243-2014.

Tang, H., Armston, J., Hancock, S., Marselis, S., Goetz, S., Dubayah, R., 2019. Characterizing global forest canopy cover distribution using spaceborne lidar. Remote Sens. Environ. 231, 111262 https://doi.org/10.1016/J.RSE.2019.111262.

Tao, S., Guo, Q., Li, C., Wang, Z., Fang, J., 2016. Global patterns and determinants of forest canopy height. Ecology 97, 3265–3270. https://doi.org/10.1002/ecy.1580.

Tedim, F., Leone, V., Amraoui, M., Bouillon, C., Coughlan, M.R., Delogu, G.M., Fernandes, P.M., Ferreira, C., McCaffrey, S., McGee, T.K., Parente, J., Paton, D., Pereira, M.G., Ribeiro, L.M., Viegas, D.X., Xanthopoulos, G., 2018. Defining extreme wildfire events: difficulties, challenges, and impacts. Fire 1, 9. https://doi.org/10.3390/FIRE1010009.

Tijerín-Triviño, J., Moreno-Fernández, D., Zavala, M.A., Astigarraga, J., García, M., 2022. Identifying forest structural types along an aridity gradient in peninsular Spain: integrating low-density LiDAR, Forest inventory, and aridity index. Remote Sens. 14, 235. https://doi.org/10.3390/RS14010235/S1.

U.S. Geological Survey, 2006. Shuttle Radar Topography Mission, 30 Arc Second product, version 2.0. Glob. L. Cover Facil. Univ. Md., Coll. Park. Md.

USGS, 2022. United States Geological Service. NASA.

van Wees, D., van der Werf, G.R., Randerson, J.T., Andela, N., Chen, Y., Morton, D.C., 2021. The role of fire in global forest loss dynamics. Glob. Chang. Biol. 27, 2377–2391. https://doi.org/10.1111/GCB.15591.

Villaescusa, R., Díaz, R., 1998. Segundo Inventario Forestal Nacional (1986–1996). Madrid.

Voulgarakis, A., Field, R.D., 2015. Fire influences on atmospheric composition, air quality and climate. Curr. Pollut. Rep. 1, 70–81. https://doi.org/10.1007/S40726-015-0007-Z/FIGURES/1.

Wang, C., Elmore, A.J., Numata, I., Cochrane, M.A., Shaogang, L., Huang, J., Zhao, Y., Li, Y., 2022. Factors affecting relative height and ground elevation estimations of GEDI among forest types across the conterminous USA. GISci. Remote Sens. 59, 975–999. https://doi.org/10.1080/15481603.2022.2085354.

Zanaga, D., Van De Kerchove, R., De Keersmaecker, W., Souverijns, N., Brockmann, C., Quast, R., Wevers, J., Grosu, A., Paccini, A., Vergnaud, S., Cartus, O., Santoro, M., Fritz, S., Georgieva, I., Lesiv, M., Carter, S., Herold, M., Li, L., Tsendbazar, N.E., Ramoino, F., Arino, O., 2021. ESA WorldCover 10 m 2020 v100. Zenodo. https://doi.org/10.5281/zenodo.5571936.

Zhang, J., Nielsen, S.E., Mao, L., Chen, S., Svenning, J.-C., 2016. Regional and historical factors supplement current climate in shaping global forest canopy height. J. Ecol. 104, 469–478. https://doi.org/10.1111/1365-2745.12510.

Zheng, B., Ciais, P., Chevallier, F., Yang, H., Canadell, J.G., Chen, Y., van der Velde, I.R., Aben, I., Chuvieco, E., Davis, S.J., Deeter, M., Hong, C., Kong, Y., Li, Haiyan, Li, Hui, Lin, X., He, K., Zhang, Q., 2023. Record-high CO2 emissions from boreal fires in 2021. Science (80-.). 379, 912–917. https://doi.org/10.1126/science.ade0805.

Zuur, A.F., Ieno, E.N., Walker, N., Saveliev, A.A., Smith, G.M., 2009. Mixed effects models and extensions in ecology with R. In: Statistics for Biology and Health. Springer New York, New York, NY. https://doi.org/10.1007/978-0-387-87458-6.